



FIVE

Improving Your Aerobic Fitness

Consider how you can feel breathless when going up a flight of stairs quickly—your body is showing the need for oxygen. “Aerobic” means “with oxygen,” and aerobic fitness, otherwise referred to as cardiorespiratory endurance, pertains to how well your body is able to take in oxygen and put that oxygen to use. Activities that involve large-muscle groups engaged in dynamic movement for prolonged periods of time are considered aerobic (2, 6). Your cardiovascular system (heart and blood vessels) and your respiratory system (lungs and air passages) work together during longer-duration activities to supply working muscles and organs with the oxygen they need. Examples of aerobic activities include walking, jogging, running, cycling, swimming, dancing, hiking, and team sports such as basketball and soccer.

Health and Fitness Benefits of Aerobic Activity

Regular and consistent aerobic activity improves your cardiorespiratory endurance. In other words, your heart, blood vessels, and lungs benefit from working harder than normal. Exercise improves your cardiorespiratory function by increasing the activity of these organ systems above what they experience at rest. Over time, your body adapts to these stresses and your fitness improves (2, 6).

Cardiorespiratory endurance is an important aspect of health for a number of reasons (2, 6, 8):

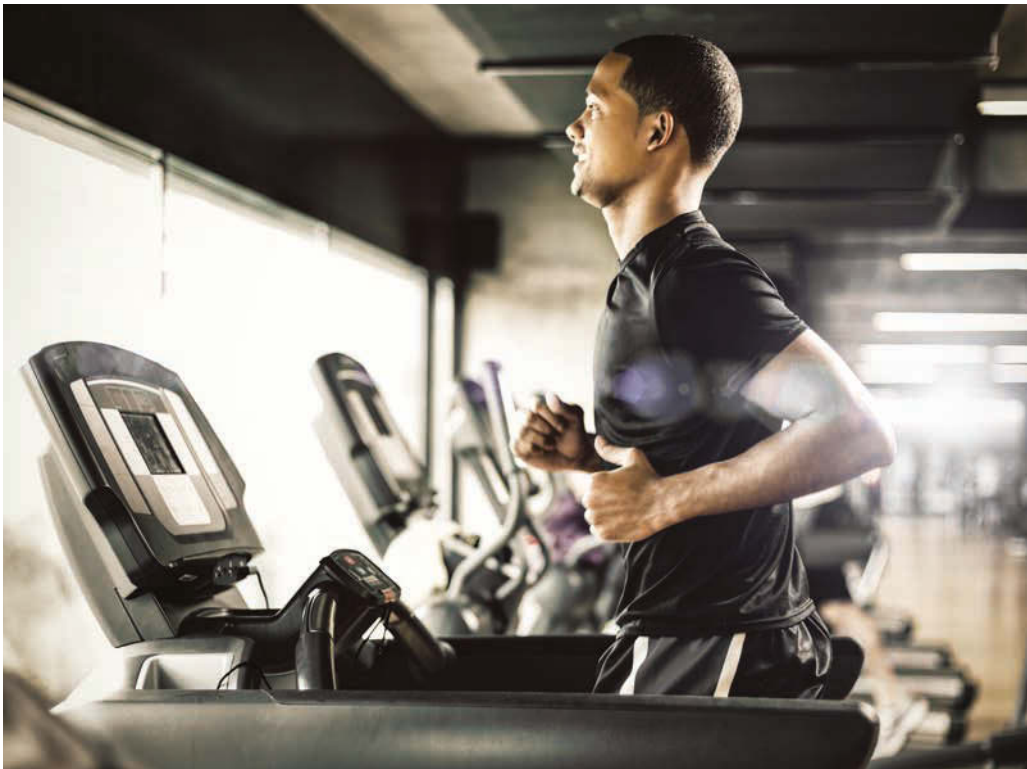
- Better cardiorespiratory endurance typically leads to higher levels of routine physical activity as you go about your day-to-day life. This in turn provides additional health benefits.
- Low levels of cardiorespiratory fitness are associated with higher risk of premature death from all causes, and specifically from cardiovascular disease. To look at this from a more positive perspective, increases in cardiorespiratory fitness are associated with a decreased risk of death from all causes.

- Aerobic fitness is an important foundation that allows you to engage in activities of daily living with greater ease.
- Increases in cardiorespiratory endurance allow you to more fully participate in recreational and sport activities.
- Aerobic activities that promote cardiorespiratory endurance also burn a relatively large number of calories and thus help to maintain appropriate body weight.

This is not an exhaustive list but does demonstrate the wide-ranging benefits of aerobic exercise for health as well as fitness.

Aerobic Fitness Assessments

Assessing aerobic fitness can provide helpful insights on your current status as well as on the progress you are making in your exercise program over time. Before engaging in any active measurement, be sure to complete the preparticipation screening steps (including follow-up with your health care provider if indicated) outlined in chapter 2. This process is intended to help verify your readiness for exercise testing and future physical activity. If you are already currently active and have no cardiovascular, metabolic, or renal disease, then you can consider any of the assessments described in the following section. If you have not been regularly active, or if you have noted any medical condition warranting lower exercise intensity or have any activity restrictions recommended by your health care provider, then select less intense assessments and



Aerobic exercise improves cardiorespiratory endurance.

consider consulting with your health care provider to ensure you are ready for exercise testing. Additional considerations are provided for each assessment described in the following sections.

Assessing Heart Rate

Perhaps the simplest fitness assessment reflecting aerobic fitness is heart rate, which is reported in beats per minute. Heart rate naturally increases during exercise. The higher the intensity, the faster your heart must beat to bring oxygen and nutrients to your working muscles. As you gain fitness, however, your heart rate will be lower at rest as well as in response to a given level of exercise. As a result of aerobic training, the heart becomes a better pump. Your heart can now do the same job while beating more slowly because it is able to push out more blood with each heartbeat. This is evidence of your body adapting to the exercise and improvement in your cardiorespiratory fitness.

You can determine your heart rate by finding a location on your body where an artery (a blood vessel carrying blood from the heart to the rest of the body) is close to the surface of the skin so you can feel your pulse, which is the slight surge in blood flow that occurs when the heart contracts. Common locations are the radial artery in the wrist and the carotid artery in the neck (see figure 5.1). Use the tips of your middle and index fingers to feel your pulse. If you use the carotid, be sure to keep the pressure light. Too much pressure at this location can alter your heart rate artificially.

Resting heart rate can be determined first thing in the morning or when you have been seated, relaxed, and inactive for a period of time. To measure resting heart rate you will need a timing device that displays time in seconds. Locate one of the arteries just described, and simply count the number of beats (pulses) you feel for 1 minute.

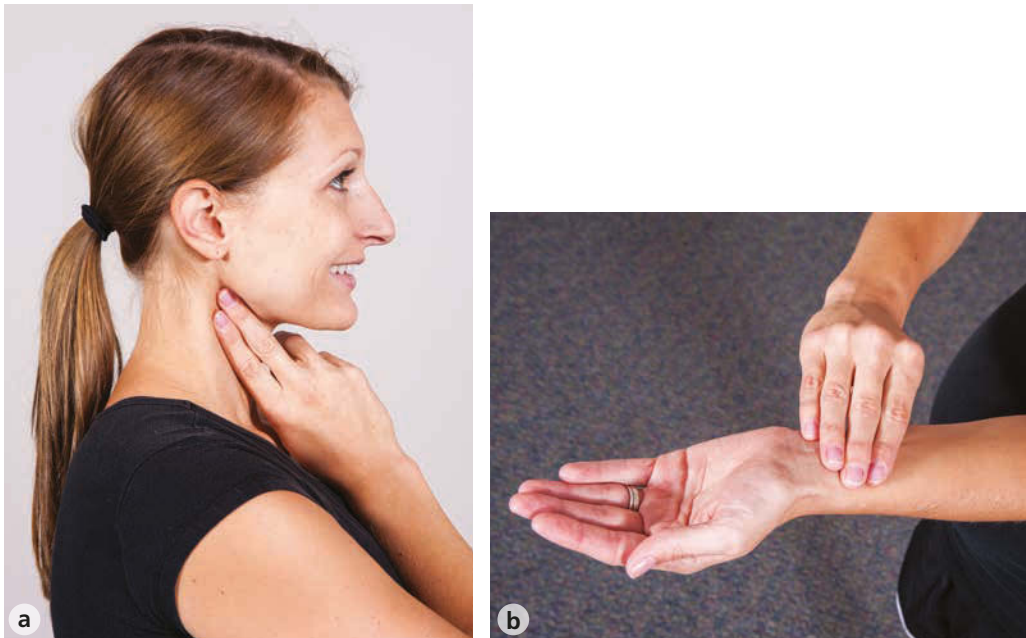


FIGURE 5.1 (a) Carotid and (b) radial artery pulse locations.

Q&A**What is a typical resting heart rate for an adult?**

For most adults, the number is between 60 and 100, but if your heart rate is lower than 60 or higher than 100 after multiple resting measurements, you should mention this to your health care provider.

Exercise heart rate is just as easy to measure as resting heart rate, but because heart rate steadily returns to a resting rate once you stop physical activity, finding your pulse and beginning your count immediately upon stopping is important. Take your pulse for 15 seconds and multiply the resulting number by 4. The answer is your exercise heart rate in beats per minute.



FIGURE 5.2 Heart rate monitor.

If manually taking your pulse is too difficult, consider making an investment in a heart rate monitor (an example is shown in figure 5.2). A heart rate monitor allows for a constant real-time readout of your heart rate by way of a transmitter (worn around the chest) that electronically communicates with a receiver that looks like a wristwatch. Heart rate is displayed on the receiver in beats per minute. The cost of heart rate monitors varies widely depending on their features (e.g., programmable heart rate zones, memory features to download to a computer after a workout, timekeeping functions). The simplest models that display only heart rate typically cost around \$25. They are very durable and allow for easy checks of your heart rate during exercise.

Estimating Aerobic Fitness Level

Aerobic fitness is typically assessed by looking at maximal oxygen consumption, also called $\dot{V}O_{2\max}$. $\dot{V}O_{2\max}$ is a marker of your body's ability to take in and use oxygen. The higher this value is, the better your aerobic fitness is (2). Complex laboratory tests can most precisely determine your $\dot{V}O_{2\max}$, but you can get a reasonable estimate from simple tests such as the Rockport One-Mile Walking Test or the 1.5-mile run test, both of which are described in this section. Other assessments are available for older people (e.g., 6-minute walk test) as well as younger individuals. For youth, a shorter-distance run test is often used (see One-Mile Run Test for Youth later in this chapter) (3).

Select one of these tests based on your current health status, as well as physical activity and perceived fitness level. The walking test is more appropriate if you are planning to begin an exercise program after a period of inactivity or currently engage in moderate levels of exercise. If you are healthy and more active, the run test is

another option. Each test and the associated calculations produce an estimation of your aerobic capacity. Use that result and the numbers provided in table 5.1 to determine your fitness level by age and sex (2).

TABLE 5.1 Fitness Levels for Aerobic Capacity* in Males and Females

Males	Age				
	20 to 29	30 to 39	40 to 49	50 to 59	60 to 69
Superior	66.3 or higher	59.8 or higher	55.6 or higher	50.7 or higher	43.0 or higher
Excellent	57.1 to 66.2	51.6 to 59.7	46.7 to 55.5	41.2 to 50.6	36.1 to 42.9
Good	50.2 to 57.0	45.2 to 51.5	40.3 to 46.6	35.1 to 41.1	30.5 to 36.0
Fair	44.9 to 50.1	39.6 to 45.1	35.7 to 40.2	30.7 to 35.0	26.6 to 30.4
Poor	38.1 to 44.8	34.1 to 39.5	30.5 to 35.6	26.1 to 30.6	22.4 to 26.5
Very poor	38.0 or lower	34.0 or lower	30.4 or lower	26.0 or lower	22.3 or lower

Females	Age				
	20 to 29	30 to 39	40 to 49	50 to 59	60 to 69
Superior	56.0 or higher	45.8 or higher	41.7 or higher	35.9 or higher	29.4 or higher
Excellent	46.5 to 55.9	37.5 to 45.6	34.0 to 41.6	28.6 to 35.8	24.6 to 29.3
Good	40.6 to 46.4	32.2 to 37.4	28.7 to 39.9	25.2 to 28.5	21.2 to 24.5
Fair	34.6 to 40.5	28.2 to 32.1	24.9 to 28.6	21.8 to 25.1	18.9 to 21.1
Poor	28.6 to 34.5	24.1 to 28.1	21.3 to 24.8	19.1 to 21.7	16.5 to 18.8
Very poor	28.5 or lower	24.0 or lower	21.2 or lower	19.0 or lower	16.4 or lower

*Aerobic capacity or $\dot{V}O_{2\max}$ expressed in $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$.

Adapted by permission from American College of Sports Medicine, 2018.

Rockport One-Mile Walking Test

The Rockport One-Mile Walking Test is a way to estimate $\dot{V}O_{2\max}$ (2). To complete this test, you should have the ability to walk 1 mile continuously. Choose a day without windy weather for testing. Ideally, you should perform the One-Mile Walking Test using an outdoor or indoor running track so that you can be certain that the distance you walk is no more or less than 1 mile. A standard quarter-mile track would be ideal (four laps on the inside lane), but many tracks are metric. If you are on a 400-meter track, then you will need to complete four laps on the inside lane plus an additional 9.3 meters (equal to approximately 31 ft). If a track is not available, any measured course will work as long as the surface is smooth and the course is flat. Grab a comfortable pair of shoes and a stopwatch. Walk the course as rapidly as you can without jogging or running, and record the time it takes for you to complete the mile. You also need to take your pulse as previously described immediately after you complete the mile walk.

Computing your results from the Rockport One-Mile Walking Test takes a bit of work, but the math is very simple when you plug results into one of the formulas shown here (numbers in bold are constant in the equations and thus are predetermined):

Males

139.150

Minus (**0.1692** × ____ weight in kilograms)

Minus (**0.3877** × ____ age in years)

Minus (**3.2649** × ____ time in minutes)

Minus (**0.1565** × ____ heart rate in beats per minute)

= ____ Aerobic capacity

Females

132.835

Minus (**0.1692** × ____ weight in kilograms)

Minus (**0.3877** × ____ age in years)

Minus (**3.2649** × ____ time in minutes)

Minus (**0.1565** × ____ heart rate in beats per minute)

= ____ Aerobic capacity

To obtain your weight in kilograms, multiply your weight in pounds by 0.454. For the time factor, you might wonder how to account for the number of seconds. For example, suppose you completed the one-mile walk in 14 minutes and 25 seconds. The 25 seconds needs to be expressed as a fraction (decimal number) of a minute. To do that, simply divide the number by 60 (because there are 60 seconds in a minute). In this case, 25 seconds would be about 0.42 of a minute, so you would use the number 14.42 in your calculation of aerobic capacity.

The answer you calculate is your aerobic capacity and refers to the amount of oxygen your body can use each minute—more specifically, the number of milliliters of oxygen your body uses per unit of body weight every minute ($\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$). The more oxygen your body can use, the better your aerobic fitness level is. Once you have determined your aerobic capacity, find your fitness classification level in table 5.1.

1.5-Mile Run Test

Just as the Rockport One-Mile Walking Test is a way to estimate aerobic capacity, so too is the 1.5-mile (2.4 km) run (2). Because of the higher intensity and longer distance of this test, it is not appropriate for beginners, anyone with symptoms of or known heart disease, or anyone with risk factors or other health concerns as determined by a health screening or a health care provider.

To perform this test, choose a day without windy weather and use an outdoor or indoor running track. If you are on a quarter-mile track, this will involve six laps in the inside lane. If you are using a 400-meter track, it will involve six laps plus an additional 14 meters (46 ft) to complete the full distance of 1.5 miles. Wear a comfortable pair of running shoes and have a stopwatch handy. Because this test requires you to run as fast as you can for 1.5 miles, you should walk a lap or two to warm up. At the track, run as rapidly as you can for 1.5 miles, timing yourself to the nearest second. For this test, there is no need to record your heart rate. This test is challenging, so be sure to walk a lap or two to cool down after completion, and rehydrate as needed afterward.

The math used to interpret your results is much simpler than that for the Rockport One-Mile Walking Test. Use the following formula to estimate your aerobic capacity:

$$\text{Aerobic capacity} = (483 \div \text{time in minutes}) + 3.5$$

As with the One-Mile Walking Test, this calculated value is an estimate of your aerobic capacity, or $\dot{V}O_{2\max}$. Because the number itself may not have much meaning, be sure to consult table 5.1 to check on your status compared to others of your age and sex (2). The higher the value, the better.

6-Minute Walk Test

Although some older adults may be comfortable completing the one-mile walk or 1.5-mile run test, another option is a 6-minute walk test (7). The 6-minute walk test could also be considered if you have been very inactive and are currently deconditioned. The test requires you to determine the distance you can walk in 6 minutes around a 50-yard (45.7 m) rectangular area (see figure 5.3 for the setup showing the number of yards walked). Focusing on a time rather than a particular distance covered allows individuals of all abilities to assess their fitness. Normal ranges for older adults are found in table 5.2. If your score is over the range listed, consider yourself above average; if your score falls short of the range, consider yourself below average (7).

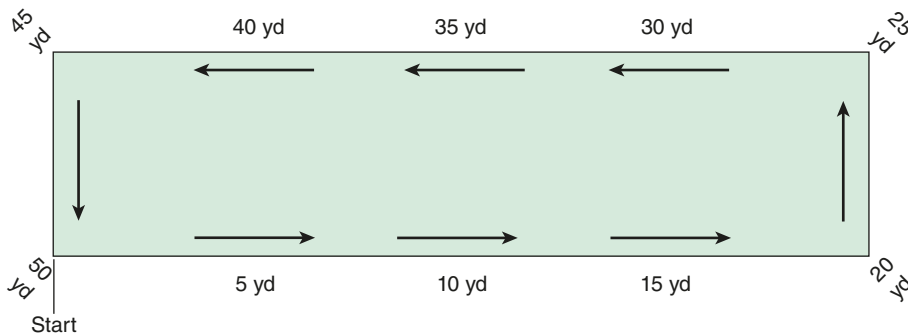


FIGURE 5.3 Setup for 6-minute walk.

Adapted by permission from R.E. Rikli and C.J. Jones, 2013, p. 76.

TABLE 5.2 **Normal Ranges for 6-Minute Walk Test for Older Adults in Yards**

	Age						
	60 to 64	65 to 69	70 to 74	75 to 79	80 to 84	85 to 89	90 to 94
Males	610 to 735	560 to 700	545 to 680	470 to 640	445 to 605	380 to 570	305 to 500
Females	545 to 660	500 to 635	480 to 615	435 to 585	385 to 540	340 to 510	275 to 440

Adapted by permission from R.E. Rikli and C.J. Jones, 2013 pp. 89, 90.

One-Mile Run Test for Youth

FitnessGram is an assessment for children that emphasizes personal fitness for health rather than comparisons among children (3). With this philosophy in mind, healthy ranges are given rather than fitness rankings. Youth are considered in the “healthy fitness zone” or “needs improvement” zone. For boys and girls between the ages of 10 and 17, a one-mile run test is used (if a child cannot run this entire distance, encourage walking at a fast pace). For the one-mile run test, the calculation to estimate aerobic capacity ($\dot{V}O_{2max}$) takes into account body mass index (BMI) as well as the time to complete the one-mile run (5).

Boys

108.94

Minus (**8.41** × ____ mile time in minutes)
Plus (**0.34** × ____ mile time in minutes × ____ mile time in minutes)
Plus (**0.21** × ____ age in years)
Minus (**0.84** × ____ BMI)
= ____ Estimated $\dot{V}O_{2max}$

Girls

108.94

Minus (**8.41** × ____ mile time in minutes)
Plus (**0.34** × ____ mile time in minutes × ____ mile time in minutes)
Minus (**0.84** × ____ BMI)
= ____ Estimated $\dot{V}O_{2max}$

Note that this estimation can be used only for run times of 13 minutes or less. If the child requires more than 13 minutes to complete the one-mile run, then simply enter “13” into the formula for the mile time. To determine BMI, see figure 18.1. See table 5.3 for the healthy fitness zone for boys and girls between the ages of 10 and 17 (4).

TABLE 5.3 **FitnessGram Standards for Aerobic Capacity Based on the One-Mile Run for Youth***

Age	Boys	Girls
10	40.2 or higher	40.2 or higher
11	40.2 or higher	40.2 or higher
12	40.3 or higher	40.1 or higher
13	41.1 or higher	39.7 or higher
14	42.5 or higher	39.4 or higher
15	43.6 or higher	39.1 or higher
16	44.1 or higher	38.9 or higher
17	44.2 or higher	38.8 or higher

*The values listed represent the healthy fitness zone and indicate that the child has a sufficient fitness level to provide important health benefits. Being below the value listed indicates a need for improvement.

Adapted by permission from The Cooper Institute, 2017, pp. 86, 87.

Aerobic Workout Components

An aerobic workout should follow a consistent pattern to optimize safety as well as enjoyment (2). You should begin with a warm-up, which is followed by the main part of the workout, called the endurance conditioning phase. The workout is then wrapped up with a cool-down. See figure 5.4 for an overview of an aerobic exercise session.

Warm-Up

A warm-up that consists of a minimum of 5 to 10 minutes of low- to moderate-level activity is essential (2). The intent of the warm-up is to increase the temperature of the muscles, thus preparing the body for the demands of the endurance conditioning phase, or main focus, of the workout. A warm-up prepares your heart, lungs, and

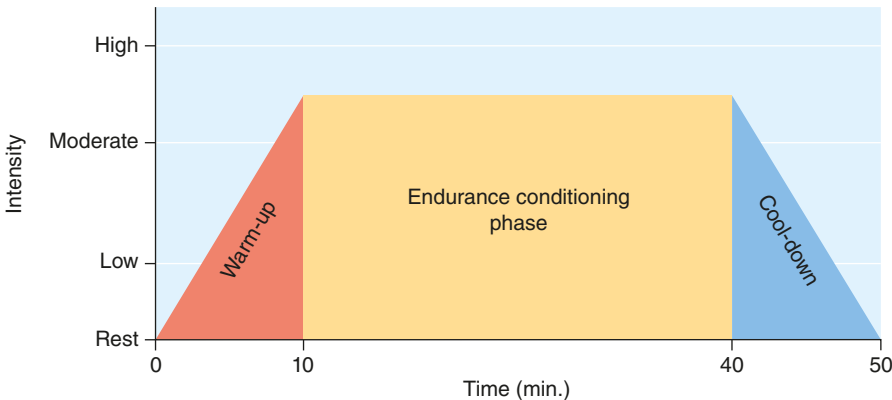


FIGURE 5.4 Overview of aerobic exercise session.

Adapted from B. Bushman and J.C. Young, 2005, p. 35.

muscles for the endurance conditioning phase of your aerobic training session (2) and may reduce the risk of injury (6). Think of the warm-up as an on-ramp to a freeway. The on-ramp gives you time to bring your vehicle up to the speed of traffic to avoid an accident. The faster the traffic is, the longer the on-ramp should be. In the same way, your warm-up should be longer if the intensity of the conditioning phase is high.

Warm-up activities may include some light calisthenics or lower-level activities similar to what you will be including in the conditioning phase. For example, if your program includes brisk walking for the conditioning phase, then the warm-up could include slower-paced walking. If the conditioning phase includes a more intense activity such as running, then jogging would be appropriate in the warm-up. The point is to gradually increase the intensity from resting levels to the intensity you plan for the conditioning phase.

Endurance Conditioning Phase

To continue with the freeway analogy, the endurance conditioning phase is the freeway itself—the main focus of your journey. The conditioning phase for aerobic activity is guided by the FITT-VP principle, which stands for frequency, intensity, time, type, volume, and progression (2). Frequency refers to the number of days per week you set aside time for exercise. Intensity reflects how hard you are working when exercising. Time simply refers to the duration you are active, on a daily or weekly basis. Type, or exercise mode, focuses on activities that involve large-muscle groups to improve cardiorespiratory fitness. Volume reflects the total amount of exercise and may be expressed in the number of calories burned. Progression refers to the manner in which the program is advanced over time as your fitness level improves.

Although FITT-VP nicely summarizes the conditioning phase, you will also want to add an “E”—the E stands for enjoyment. All the recommendations and information in the world mean little if you do not stick with your exercise program. Understanding the benefits of an exercise program (as outlined in chapter 1) may keep you active, but considering the time commitment you are making, you should also be sure you are having some fun. Suggestions for keeping exercise enjoyable are found later in this chapter. First, consider the nuts and bolts of an aerobic exercise program.

Frequency

The recommended frequency of aerobic exercise is three to five days per week. How many days you exercise depends on your goals and the intensity that is most appropriate for you. Although as few as a couple of days per week of activity can provide benefits, regular physical activity provides more benefits and has a lower risk of musculoskeletal injury than sporadic activity (2, 6). You will need as few as three days per week if you are engaging in vigorous activity, but at least five days per week is recommended if you plan on moderate-intensity activity. For example, if you enjoy running (a vigorous activity), three days per week will provide you with health and fitness benefits. However, if you plan on a walking program (a moderate-intensity activity), then at least five days per week would be better. If you enjoy mixing types and intensities of activity, then a weekly combination of three to five days of moderate and vigorous activity is recommended (2, 8). For example, you may walk a couple days per week and jog on another couple days. This would be considered two days per week of moderate activity (i.e., walking) and two days per week of vigorous activity (i.e., jogging), allowing you to meet the recommended amount of physical activity.

Intensity

As the intensity of activity increases, so do the potential health benefits. To promote health and fitness benefits, your exercise must place some stress on your cardiorespiratory system. In other words, you should notice an increase in your heart rate and breathing. When speaking of intensity, fitness professionals generally use the terms moderate and vigorous (2, 8). To help visualize this, consider moderate-intensity activity to be equivalent to brisk walking and vigorous-intensity activity to be equivalent to jogging or running (8).

A variety of simple methods are available to help you quantify the intensity of your exercise bout. One method is to monitor your relative level of effort. Although this is subjective (i.e., you determine how easy or hard you are exercising), a numerical scale can help guide you to appropriate levels of activity. The U.S. Department of Health and Human Services' *Physical Activity Guidelines for Americans* suggests a scale of 0 to 10. Sitting at rest is 0, and your highest effort level possible is 10 (8). Moderate-intensity activity is a 5 or 6 on this effort scale. Vigorous-intensity activity is at a level of 7 or 8. This method allows you to individualize your exercise based on your current level of cardiorespiratory fitness (8). For an example of applying this scale, see figure 5.5.

Another method, called the talk test, can also be used to establish exercise intensity (2). If you are working at an intensity that increases breathing rate but still allows you to speak without gasping for breath between words, you are likely exercising at a moderate intensity. The goal would be to exercise to the point at which speech would start to become more difficult. The *Physical Activity Guidelines for Americans* suggests that moderate-intensity activity allows you to talk but not to sing, whereas more vigorous activity results in an inability to say more than a few words without pausing for a breath (8).

Heart rate monitoring can also be helpful for determining your intensity level, although it is a bit more technical than the subjective measures of effort level and the talk test. Maximal heart rate can be estimated by subtracting your age in years from 220 (2). Thus for a 40-year-old, estimated maximal heart rate would be 180 beats per minute (i.e., $220 - 40 = 180$). You will not be exercising at maximal heart rate, but rather at a percentage of that value; the percentage will depend on your target level of intensity (2). Multiply your estimated maximal heart rate by the activity factor from table 5.4 to determine your target heart rate.

$$\text{___ estimated maximal heart rate} \times \text{___ activity factor} \\ = \text{target exercise heart rate in beats per minute}$$



FIGURE 5.5 Sample scale for where activities fall within the various intensity levels.

TABLE 5.4 **Heart Rate Intensity Guidelines**

Intensity level	Percentage of maximal heart rate	Activity factor*
Very light	~55%	0.55
Light	~60%	0.60
Moderate	~70%	0.70
Vigorous	~85%	0.85

*Multiply activity factor by maximal heart rate to determine target heart rate.

Adapted by permission from American College of Sports Medicine, 2018.

Note that your heart rate can also be influenced by environmental conditions (e.g., hot, humid environments) as well as medications (e.g., beta-blockers used for migraines and heart disease can lower heart rate). The calculated value should be used in conjunction with relative perception of effort or the talk test (2). You can adjust your workload up or down depending on your perception of effort on a given day.

Recognize, too, that you can vary your intensity during the conditioning phase. Athletes often use interval training, which includes some time at higher intensity followed by lower-intensity exercise. This provides a unique stress on the body that translates into improved aerobic fitness. This principle can be used for general exercise programs as well (2). For example, if you are just beginning to exercise, you could include a few minutes at a faster walking pace within your conditioning phase. Alternating between lower and higher intensity provides variety as well as a stimulus to improve your aerobic capacity, no matter your current level of fitness.

Time

The duration of each of your exercise sessions is determined by the amount of time you are able to commit as well as your current fitness status. If you are a beginner, don’t worry about some arbitrary time goal; rather, find an activity that you can do continuously for 10 minutes. Increase the duration of the exercise session as it becomes easier to complete. Add a couple of minutes per session until you reach about 30 minutes of aerobic exercise per day. Depending on your initial fitness level, this may take weeks or even a month or more. The key is to keep going and make progress.

If you have already been doing some exercise (or have now built up to 30 minutes of continuous activity) and feel comfortable with moderate-intensity activity for this length of time, decide whether you want to maintain your current intensity and go for a bit longer, or if you want to begin to increase the intensity. Time and intensity are like a teeter-totter. When you increase intensity, you generally decrease the length of the session. If you decrease intensity, you will need to increase the time you spend exercising to achieve full health benefits. A general rule of thumb from the *Physical Activity Guidelines for Americans* is that 1 minute of vigorous-intensity activity can be counted as the same as 2 minutes of moderate-intensity activity (8). For example, a 15-minute run would provide the same health benefit as a 30-minute walk.

Labels are difficult to apply universally, but table 5.5 provides some terminology related to activity status that was introduced in chapter 2. For the purposes of this book, beginners are those who currently have limited activity. As you can see in the table, beginners are focusing on very light to light activity and build up to 100 to 150 minutes per week of light to moderate activity. The intermediate level of activity reflects

TABLE 5.5 **Activity Status and Aerobic Training Focus**

Activity status	Aerobic training focus
Beginner (inactive with no or minimal physical activity and thus relatively deconditioned)	<i>No prior activity:</i> Focus is on very light to light activity for 20 to 30 min over the course of the day. Accumulating time in 10-min bouts is an option. Overall, your target is 60 to 100 min per week. <i>Some prior activity</i> (i.e., once you have met the target level of 60-100 min per week): Focus is on light- to moderate-level activity for 30 to 45 min per day. Accumulating time in 10-min bouts is an option. Overall, your target is 100 to 150 min per week.
Intermediate (somewhat active but overall only moderately conditioned)	<i>Some activity</i> (fair to average fitness): Focus is on moderate activity for 30 to 60 min per day. Overall, your target is 150 to 250 min per week.
Established (regularly engaging in moderate to vigorous exercise)	<i>Regular exerciser</i> (moderate to vigorous): Focus is on moderate- to vigorous-intensity activity for 30 to 90 min per day. Overall, your target is 150 to 300 min per week (duration depends on intensity).

people who are somewhat active and are moderately conditioned. The focus at this stage is increasing moderate-intensity aerobic activity to 150 to 250 minutes per week. Typically, people at this level are of fair to average fitness levels. Established exercisers are those who have been engaged in regular exercise for at least six months. Fitness levels vary according to genetic potential as well as personal fitness goals. Typically, established exercisers have average to excellent aerobic fitness.

The *Physical Activity Guidelines for Americans* recommends working toward a minimum of 150 minutes per week of moderate-intensity activity, or 75 minutes per week of vigorous-intensity activity (8). If you are already physically active at this level, then consider increasing your activity to gain additional health and fitness benefits. For you, a new target of 300 minutes per week of moderate-intensity activity, or 150 minutes per week of vigorous-intensity activity, would be a potential goal (8).

Q&A

How can interval training be designed to provide variety in an exercise program?

Interval training occurs when exercise intensity varies during an exercise session. This type of training provides many options, as you can change the number, duration, and intensity of various phases of the exercise session (2). For example, you could engage in moderate exercise at a level 5 (on the 10-point exertion scale) for 2 minutes followed by 3 minutes of vigorous exercise at a level 7 (on the 10-point exertion scale) and repeat that sequence four times for a total of 20 minutes for the exercise session. To provide variety, you can change the time spent or the intensity of each of the different intervals. For example, 2 minutes at level 6 followed by 2 minutes at level 8 could be repeated five times for a total of 20 minutes for the exercise session. The options are almost limitless and can be individualized based on your current health and fitness status (2).

Type or Mode

Aerobic activities are grouped into four categories along with recommendations on who would most appropriately engage in the given activity (see table 5.6) (2). Exercises in group A are recommended for everyone because they are relatively simple activities that can be started at a low level of effort. Group B activities are more vigorous and thus are most appropriate if you already have a good fitness base (i.e., you have been exercising regularly and have determined your fitness level to be at least in the fair to average range). Group C activities are those that have a definite skill component and thus may require some learning before being used as a fitness tool. Group D activities are recreational and, because intensity varies depending on the situation, are best reserved for people who are regularly active and have a good fitness base. Do not consider these groupings progressive (e.g., that group C activities are better than group B activities), but rather as a way to classify various aerobic exercises.

Volume

The concept of volume reflects a summary or overall amount of activity. One way to provide a summary of your aerobic exercise is to determine the calories you use when engaging in your aerobic activities each week. When considering the activity recommendations in the *Physical Activity Guidelines for Americans*, a reasonable target is at least 1,000 calories per week (2). Calculating calories burned can be helpful when you are interested in losing weight, but it is also a great way to pull together the four parts of your aerobic exercise prescription—frequency, intensity, time, and type of activity—into one number. Whether you do the same activity each day or change it up, you still can take a look at your weekly total to ensure that you are on track with just a few calculations.

TABLE 5.6 Aerobic Exercise Groupings

Exercise group	Group characteristics	Recommended participants	Examples
A	Endurance activities that can be done with minimal skill and with minimal fitness	Everyone	Walking, easy bicycling, slow dancing
B	Endurance activities that are more vigorous but can be done with minimal skill	Because of the higher intensity, adults who are regularly active and have at least an average level of fitness would be best suited.	Jogging, running, spinning, elliptical exercise, fast dancing
C	Endurance activities that require a certain level of skill to perform	Assuming that a skill level has been achieved, people should have at least an average level of fitness to be suited for these activities.	Swimming, cross-country skiing, skating
D	Recreational sports	Because of the changing exertion level due to competition or terrain, people should have at least an average level of fitness.	Basketball, tennis, soccer, downhill skiing, hiking

Adapted by permission from American College of Sports Medicine, 2018.

To keep things simple, researchers have created a unit of measure called a metabolic equivalent, or MET. A MET is equal to the oxygen cost at rest (i.e., 1 MET = resting level = 3.5 milliliters of oxygen per kilogram body weight per minute). Multiples of a MET are then applied to various activities. For example, walking at 3.5 miles per hour (5.6 km/h) is equal to 4 METs. In other words, you are working four times harder when walking at 3.5 miles per hour than you are when seated in a resting position. Metabolic equivalent values have been determined for a wide variety of activities (see table 5.7 for some examples of basic activities) (1).

Once you know the MET value for a given exercise, you can estimate how many calories you burned per minute by inserting that value into the following formula (numbers in bold are constants—in other words, they do not change):

$$\text{___ MET value of activity} \times \mathbf{3.5} \times \text{___ body weight in kg} \div \mathbf{200} \\ = \text{___ calories burned per minute}$$

Insert the MET value for the activity and then your body weight (to convert from pounds to kilograms, multiply your weight in pounds by 0.454 to determine your weight in kilograms). For an example on how this can be used, see Checking Volume of Aerobic Exercise.

TABLE 5.7 MET Values for Selected Activities*

Activity	MET value
Bicycling outdoors, <10.0 mph (16 km/h), leisure riding	4.0
Bicycling outdoors, 10.0 to 11.9 mph (16 to 19.2 km/h)	6.8
Bicycling outdoors, 12.0 to 13.9 mph (19.2 to 22.4 km/h)	8.0
Bicycling outdoors, 14.0 to 15.9 mph (22.5 to 25.6 km/h)	10.0
Biking, stationary, 30 to 50 W, very light to light effort	3.5
Biking, stationary, 90 to 100 W, moderate to vigorous effort	6.8
Biking, stationary, 101 to 160 W, vigorous effort	8.8
Biking, stationary, 161 to 200 W, vigorous effort	11.0
Running, 5 mph (8 km/h)	8.3
Running, 6 mph (9.7 km/h)	9.8
Running, 7 mph (11.3 km/h)	11.0
Running, 8 mph (12.9 km/h)	11.8
Swimming laps, freestyle, light to moderate effort	5.8
Swimming laps, sidestroke	7.0
Swimming laps, backstroke	9.5
Swimming laps, breaststroke	10.3
Swimming laps, butterfly	13.8
Walking, 2.0 mph (3.2 km/h)	2.8
Walking, 2.5 mph (4 km/h)	3.0
Walking, 3.5 mph (5.6 km/h)	4.3
Walking, 4.5 mph (7.2 km/h)	7.0

*For a comprehensive list of activities and MET values, see <http://prevention.sph.sc.edu/tools/compendium.htm>.

Source: B.E. Ainsworth, W.L. Haskell, S.D. Herrmann, et al.

Progression

Progression is how an exercise program is advanced over time. Many factors must be considered, including current health and fitness status, training responses, and goals (2). The key is gradual progression rather than making abrupt or significant changes in one of the FITT components. If you are just starting, to optimize safety and avoid injury, the recommendation is “start low and go slow” (2). Table 5.5 reflects this concept of slowly increasing the volume of exercise. Rather than increasing frequency, intensity, and duration all at once, you want to gradually introduce changes. For example, initially, you may simply increase the time spent in activity. As you adjust to this level of activity, you may then want to cut back the time a bit and increase the intensity slightly. Reflect on the overall volume of exercise to help make sure your progression is gradual. As you make adjustments to your program, give yourself time at a particular volume of activity to ensure you are able to maintain this new level before trying to move forward.

Cool-Down

The cool-down should consist of a minimum of 5 to 10 minutes of low- to moderate-level activity (2). The cool-down provides an opportunity for body systems to gradually return to preexercise levels. A cool-down is recommended to allow the heart to slow down in a controlled manner, thus avoiding negative changes in heart rhythm. In addition, if you stop your activity too abruptly, blood that was circulating to the working muscles can pool in your legs, resulting in a drop in blood pressure. A cool-down also helps to gradually decrease body temperature, which naturally increased



Walking and jogging are common aerobic activities.

Checking Volume of Aerobic Exercise

To compare two programs—one focused on walking and the other on jogging—take a look at the MET values to help you examine how intensity influences the number of calories burned.

Walking program: walking 3.5 miles per hour (5.6 km/h) for 50 minutes

Jogging program: running at 5 miles per hour (8 km/h) for 25 minutes

For this example, the calculations are done for a 150-pound (68.1 kg) person. The MET values for each activity are found in table 5.7.

Walking at 3.5 miles per hour (5.6 km/h) is equal to 4.3 METs, so using the formula provided previously, a 50-minute workout burns about 255 calories (determined by multiplying 5.1 calories per minute by the workout duration of 50 minutes), as follows:

$$(4.3 \text{ METs} \times 3.5 \times 68.1 \text{ kg}) \div 200 = 5.1 \text{ calories per minute}$$

Running at 5 miles per hour (8 km/h) is equal to 8.3 METs, so using the formula provided previously, a 25-minute workout would burn 248 calories (determined by multiplying 9.9 calories per minute by the workout duration of 25 minutes), as follows:

$$(8.3 \text{ METs} \times 3.5 \times 68.1 \text{ kg}) \div 200 = 9.9 \text{ calories per minute}$$

The two workouts burn approximately the same number of calories. Thus even though the activities are very different, the overall volume (which accounts for the type, duration, and intensity) is similar.

during the endurance phase. Activities included in a cool-down are similar to those in the warm-up, but the intensity needs to gradually diminish toward resting levels (2).

A proper cool-down is driven by both practical issues (e.g., avoiding fainting from a drop in blood pressure) and safety issues (e.g., avoiding negative changes in heart rhythm). The cool-down is like a freeway off-ramp. When shifting from freeway speeds to those appropriate on city streets, time is needed for an adjustment. In a similar way, the cool-down allows the body to adjust back toward normal resting levels. The higher the intensity of your conditioning phase, the longer your cool-down should be.

Your Aerobic Program

If you are just getting started with your exercise program, be sure to complete the preparticipation screening process found in chapter 2 (2). This screening can help you determine whether you should visit your health care provider before starting an exercise program. Of course, regardless of the outcome, consulting with your personal health care provider is always appropriate. In addition, you need to consider your current fitness level and begin at a point suitable to your current status. Over time, with regular activity, you will progress and improve.

Your personal exercise prescription takes into account the frequency, intensity, time, and type of activity. Take walking, for example, which is the most commonly reported exercise and is a great activity for the start of an exercise program (walking is a group A activity as shown in table 5.6). Figure 5.6 shows an example of a progressive walking and jogging program. You can determine where to enter into the exercise progression based on your current level of fitness.

FIGURE 5.6

Sample walking and jogging program.

Stage	Time point	Warm-up	Workout*	Cool-down
Beginner	Initial week	Slow, easy walking pace for a couple of minutes	Walk at a pace that involves a light level of exertion (level 3 or 4) for 10 min at least twice a day for a total of 20 min each day (three days per week). Your weekly total should be 60 min.	Slow, easy walking pace for a couple of minutes
	Progression	Slow, easy walking pace for 5 min	Each week add 10 min to your weekly total until you reach 100 min of activity (e.g., 20 min five days per week). Stay at this duration and increase your intensity over the next couple of weeks from light (level 3 or 4) to moderate (level 5 or 6). Once you are comfortable with this time and intensity for a couple of weeks, continue to add 10 to 15 min per week until you reach 150 min.	Slow, easy walking pace for 5 min
	Final week	Easy walking pace for 5 to 10 min	Walk at a pace that involves a moderate level of exertion (level 5 or 6) for 30 to 60 min (three to five days per week). Your weekly total should be 150 min.	Easy walking pace for 5 to 10 min
Intermediate	Initial week	Easy walking pace for 5 to 10 min	Walk at a pace that feels moderate (level 5 or 6) for 30 to 60 min (three to five days per week). Your weekly total should be 150 min.	Easy walking pace for 5 to 10 min
	Progression	Easy walking pace for 5 to 10 min	Continue to increase exercise duration by 10 to 15 min per week to approach 200 to 250 min of moderate activity accumulated on a weekly basis. Another option is to introduce a slightly more vigorous activity, such as jogging, realizing that the time needed will be less (typically 2 min of moderate activity equals 1 min of vigorous activity).	Easy walking pace for 5 to 10 min
	Final week	Easy walking pace for 5 to 10 min	Walk at a pace that feels moderate (level 5 or 6) for 30 to 60 min (three to five days per week). Your weekly total should be 200 to 250 min (moderate intensity). Or: Combine moderate and vigorous walking on alternate days.	Easy walking pace for 5 to 10 min
Established	Continue, maintain	Easy walking pace for 5 to 10 min	Walk at a pace that feels moderate (level 5 or 6). Your weekly total should be 200 to 300 min (moderate intensity). Or: Jog (level 7 or 8). Your weekly total should be 100 to 150 min (vigorous intensity). Or: Combine moderate and vigorous walking on alternate days.	Easy walking pace for 5 to 10 min

*Level of exertion is on a scale of 0 to 10 (sitting at rest is 0, and your highest effort level is 10).

Once you feel comfortable with 30 minutes of continuous moderate-intensity activity, you may be interested in other activity options. Swimming, a group C activity, is another excellent aerobic activity if you have basic swimming skills or are willing to gain those skills. Follow the time and intensity progression described in figure 5.6, substituting swimming (using different strokes for variety) for walking and jogging.

Figure 5.7 provides a sample program for someone with a membership at a health club. Activities at the club, when done at a low intensity, would fall into group A, but as the person's fitness level improves, the intensity increase will likely result in a shift to group B exercise.

FIGURE 5.7

Sample cross-training program at a health club.

Stage	Time point	Warm-up	Workout*	Cool-down
Beginner	Initial week	Slow, easy walking pace for a couple of minutes	Pick one activity each day at a light level of exertion (level 3 or 4) for 10 min at least twice a day for a total of 20 min each day (three days per week). Select from walking on the treadmill or stationary biking. Your weekly total should be 60 min.	Slow, easy walking pace for a couple of minutes
	Progression	Slow, easy walking pace for 5 min	Each week add 10 min to your weekly total until you reach 100 min of activity (e.g., 20 min five days per week). Potential activities include treadmill walking, stationary biking, and using a stair climber. Stay at this duration and increase your intensity over the next couple of weeks from light (level 3 or 4) to moderate (level 5 or 6). Once you are comfortable with this time and intensity for a couple of weeks, continue to add 10 to 15 min per week until you reach 150 min.	Slow, easy walking pace for 5 min
	Final week	Easy walking pace for 5 to 10 min	Exercise at an intensity that involves a moderate level of exertion (level 5 or 6) for 30 to 60 min (three to five days per week). Activities may include treadmill walking; stationary biking; or using a stair climber, elliptical trainer, rowing machine, or Nordic ski machine. Your weekly total should be 150 min.	Easy walking pace for 5 to 10 min
Intermediate	Initial week	Easy walking pace for 5 to 10 min	Exercise at a level that feels moderate (level 5 or 6) for 30 to 60 min (three to five days per week) using a treadmill, stationary bike, stair climber, elliptical trainer, or Nordic ski machine. Your weekly total should be 150 min.	Easy walking pace for 5 to 10 min

> continued

Figure 5.7 > continued

Stage	Time point	Warm-up	Workout*	Cool-down
	Progression	Easy walking pace for 5 to 10 min	Continue to increase exercise duration by 10 to 15 min per week to approach 200 to 250 min of moderate activity accumulated on a weekly basis. Another option is to introduce slightly more vigorous activity a couple of days per week, such as jogging on the treadmill, taking a spinning class, or joining a step aerobics class, realizing that the time needed will be less (typically, 2 min of moderate activity equals 1 min of vigorous activity).	Easy walking pace for 5 to 10 min
	Final week	Easy walking pace for 5 to 10 min	Exercise at a level that feels moderate (level 5 or 6) for 30 to 60 min (three to five days per week). Your weekly total should be 200 to 250 min (moderate intensity). Or: Combine moderate and vigorous walking on alternate days.	Easy walking pace for 5 to 10 min
Established	Continue-maintain	Easy walking pace for 5 to 10 min	Exercise at an intensity that feels moderate (level 5 or 6). Your weekly total should be 200 to 300 min (moderate intensity). Or: Exercise at a higher intensity (level 7 or 8). Your weekly total should be 100 to 150 min (vigorous intensity). Or: Combine moderate and vigorous walking on alternate days.	Easy walking pace for 5 to 10 min

*Level of exertion is on a scale of 0 to 10 (sitting at rest is 0, and your highest effort level is 10).

The examples in figures 5.6 and 5.7 show a progression from beginner to established exerciser. Depending on your current status, you may be at the start of the table as a beginner or already in the established, or maintenance, phase. If you are just beginning to exercise, progress slowly and base your advancement on how your body is responding to the exercise. If you are in the established, or maintenance, phase, keep tracking your activity. Also, stay focused on the FITT-VP factors as discussed previously, and if you are becoming bored with your current activity program, consider other modes of exercise or joining an exercise group.

As you move along in your exercise journey, increase the duration (time) first; once you are comfortable with the activity at the longer session length, then consider increasing the intensity. To avoid injury, do not increase the session duration and intensity at the same time. Although placing a stress on the body is necessary for improvement, excessive overload can result in injury as well as frustration. To keep steady forward progress, refer to table 5.5 for general guidance.

In addition, as you examine the sample programs, once again consider the FITT-VP factors as discussed earlier and how each relates to your fitness goals. Don't forget about enjoyment. As you create your plan of action, consider the types of activities that you enjoy and that also are accessible to you. Joining a health club can be a great way to increase your access to a variety of activities (equipment as well as group classes). If you don't want to join a health club, you can easily find aerobic activities at no cost. Walking and running trails are becoming more common in cities; many malls open their doors early to allow walkers to use the corridors before the stores open; and your local library has many aerobic exercise videos that you can use in the privacy of your own home. To get started, you need to pick a day and take the first step—literally as well as figuratively.

Cardiorespiratory (or aerobic) fitness is important for promoting health and, in particular, is associated with a reduced risk of cardiovascular disease. An aerobic exercise session includes a warm-up, a conditioning phase, and a cool-down. The warm-up and cool-down are links between the resting state and the exercise portion of your workout. The main focus, the endurance conditioning phase, is guided by the FITT-VP principle: frequency, intensity, time, type, volume, and progression. General recommendations are as follows: three to five days per week (frequency), moderate to vigorous level of exertion (intensity), 20 to 30 minutes or more per session (time), large-muscle group activity (type of activity), total of 1,000 calories burned per week (volume), and gradual increases over time (progression). In addition, tracking aerobic fitness assessments periodically is a helpful way to determine current status and the effectiveness of your aerobic exercise program.

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SIX

Enhancing Your Muscular Fitness

Muscular fitness is a global term that includes muscular strength, endurance, and power. Muscular strength refers to the ability to lift a heavy weight one time, muscular endurance is the ability to lift a lighter load several times, and muscular power refers to ability to exert maximum effort in a very short period of time. Muscle-strengthening activities that involve all the major muscle groups are recognized as an essential component of an overall fitness program for both adults and youth (1, 6).

Just as aerobic fitness is improved by stressing the heart and lungs, muscular fitness requires a stress, or resistance, to be placed on the muscles. Resistance training (also called strength training) involves the use of a variety of activities that include free weights (barbells and dumbbells), weight machines, elastic tubing, medicine balls, stability balls, and body weight. Resistance training does not refer to one specific mode of conditioning but rather to an organized process of exercising with various types of resistance to enhance muscular fitness.

When correctly performed and sensibly progressed over time, resistance training can be a safe, effective, and enjoyable method of exercise for people of a wide range of ages, fitness levels, and health conditions (1, 20). While resistance training has been a part of sport programs for many years, public health recommendations now aim to increase participation in muscle-strengthening activities for all youth and adults (28, 33). With instruction on developing proper exercise technique and guidance on sensibly progressing the exercise program, resistance training can offer observable health and fitness value.

Health and Fitness Benefits of Resistance Training

To maintain your physical capacity, you must make a lifestyle choice to include resistance training on a regular basis. Unfortunately, physical capacity and muscle strength decrease dramatically with age in adults who do not engage in resistance training (3,

Resistance Training Terminology

Following are definitions of some common terms used in the design of a resistance training workout:

Atrophy—A reduction in muscle fiber size.

Concentric—A type of muscle action that occurs when the muscle shortens.

Eccentric—A type of muscle action that occurs when the muscle lengthens.

Hypertrophy—An enlargement in muscle fiber size.

Muscular endurance—The ability to repeat or maintain muscle contraction.

Muscular strength—The ability to exert maximal force in a single effort.

Repetition—One complete movement of an exercise.

Repetition maximum (RM)—The maximum amount of weight that can be lifted for a predetermined number of repetitions with proper exercise technique.

Set—A group of repetitions performed without stopping.

Spotter—A training partner or fitness professional who can provide assistance in case of a failed repetition.

30). Resistance training results in stronger muscles and therefore an increased capacity for force production, which is not achievable with solely aerobic-based training. Because muscles function as the engine of your body, they must be used regularly to avoid disuse atrophy (i.e., a reduction in muscle size) and age-related declines in physical performance.

You don't need to be a competitive athlete to benefit from resistance training; it is equally important from a health and fitness perspective. The benefits of resistance training include favorable changes in body composition, metabolic health, and quality of life. Resistance training activities can increase lean muscle mass, reduce body fat, fortify bone, lower blood pressure, improve blood lipid and cholesterol levels, and enhance your body's ability to use glucose (2, 12). These benefits can optimize your day-to-day functioning while limiting the development of chronic diseases such as diabetes, heart disease, and osteoporosis (30, 32). Of paramount importance, regular participation in a resistance training program can help adults preserve their muscle health to maintain independent physical functioning with advancing age (23, 27).

Skeletal muscle represents about 40 percent of one's total body weight and influences a variety of physiological processes and disease risk factors (26). The increase in muscle tissue that results from resistance exercise is accompanied by an increase in resting metabolic rate; the decrease in muscle tissue that results from a sedentary lifestyle is accompanied by a decrease in resting metabolic rate. Muscle mass declines about 5 percent each decade after age 30, and this loss can reach 10 percent per decade after age 50 (15). This gradual decrease in muscle mass and metabolism is associated with the gradual increase in body fat that typically occurs with age. Calories that were previously used by muscle tissue (now smaller as a result of disuse) are stored as fat. On the other hand, resistance training raises resting metabolic rate and results in more calories burned on a daily basis. In theory, if you resistance train and gain 2 pounds (~1 kg) of muscle mass, your resting metabolic rate should increase by about 20 calories per day (29). Thus, performing resistance training throughout your life can help you

recharge your metabolism, facilitate physical function, and maintain your health (30).

In addition to the effect of muscle on metabolism, another benefit of regular resistance training is an increase in bone mineral density that may reduce the risk of osteoporosis (8, 9, 11). On top of the direct effect of strength-building (and weight-bearing) exercises on bone, the act of muscles pulling on bones during resistance exercises may also be a potent stimulus for new bone formation in certain people. This potential benefit is of particular importance to women who are at increased risk of functional limitations as a result of age-related losses of bone mass.

Strong muscles serve as shock absorbers and balancing agents that help dissipate the repetitive landing forces from weight-bearing activities for active people and also reduce the risk of falling in older adults (3, 31). As such, a resistance training program that requires agility and balance may be the most effective way to enhance movement control and avoid injury (1, 10). Moreover, strength-building activities are particularly important for decreasing physical discomfort associated with low back pain, which is a growing health care concern (31).

Regular participation in resistance training activities that are consistent with your needs, goals, and abilities can improve muscle function, enhance quality of life, and lower the risk of premature all-cause mortality (18, 25). The health and fitness benefits are clear. You can also realize benefits linked to personal appearance. Firm, toned muscles are possible with regular resistance training. Whether you are seeking to improve in recreational or sport activities or just to look and feel better, resistance training should be part of your fitness program.

Assessments for Muscular Fitness

There is not one test of muscular fitness that is best. Rather, different tests can be used to safely and effectively assess muscular strength or muscular endurance in various age groups. This section describes several assessments that can be used.

Assessing Muscular Strength

A common assessment of muscular strength is called the one-repetition maximum (1RM), in which the goal is to lift as much weight as possible on a strength exercise with proper

Q&A

What is the typical impact of aging on muscle and metabolism?

As a person ages, decreases in muscle along with a lower resting metabolic rate result in less than optimal changes in body composition. For example, consider a 160-pound (72.6 kg) male with 15 percent body fat at age 30. He therefore has 24 pounds (10.9 kg) of fat weight and 136 pounds (61.7 kg) of lean weight, which consists of muscle, bone, blood, skin, organs, and connective tissue. If he weighs the same (160 pounds) at age 50, his body composition will have changed by about 20 pounds (9 kg)—10 pounds (4.5 kg) less lean weight and 10 pounds (4.5 kg) more fat—and he will now be 21 percent body fat. Of course, this increase in percentage of body fat and decrease in lean weight would have a negative impact on his appearance, health, and fitness.



Muscular fitness is a part of recreational and daily activities.

decimal value. Finally, divide the weight you lifted by that decimal value to estimate your 1RM on that exercise.

For example, if a 35-year-old female can lift 60 pounds (27 kg) on the chest press exercise 10 times, then she can use the following steps to estimate her 1RM:

$$10 \text{ repetitions} \times 2.5 = 25$$

$$100 - 25 = 75$$

$$75 \div 100 = 0.75$$

$$60 \text{ pounds} \div 0.75 = 80 \text{ pounds} = \text{estimated 1RM}$$

To compare her performance with others of her same age and sex, the 1RM is divided by body weight. In the previous example, if the individual's body weight is 145 pounds (66 kg), then she can complete the calculation ($80 / 145 = 0.55$) and use the result (0.55) to assess her performance with table 6.1 (and to assess lower body strength with table 6.2). Note that the ratio of weight lifted to body weight is the same whether you use pounds or kilograms. For a 35-year-old female, her upper body strength is in

technique for one repetition only. This test is time-consuming and should be performed under the supervision of a qualified fitness professional. Also, familiarization and practice sessions are critical to ensure that the test is safe and accurate (20).

Another option is to estimate your 1RM by lifting a submaximal weight multiple times. While different exercises can be used for this assessment, the use of multijoint exercises such as the leg press and chest press is common. With a few calculations you can estimate your 1RM and compare your performance to that of others of your age and sex.

First, multiply the number of repetitions you can perform on a given exercise by 2.5. Try to select a weight you can lift about 10 to 15 times with proper form (note that if you can lift the weight more than 20 times, the results will be more accurate if you rest and then repeat the test with a heavier weight). Subtract that number from 100 to determine the percentage of your theoretical 1RM. Then, divide that number by 100 to produce a

the “fair” category. With regular resistance training she will see her strength improve as she tracks her progress. A weight she could lift only 10 times will be lifted more often before fatiguing, or she will be able to lift a heavier weight for those same 10 repetitions.

TABLE 6.1 Interpretation of Upper Body Strength for Males and Females*

	Age					
	20 or younger	20 to 29	30 to 39	40 to 49	50 to 59	60+
Males						
Superior	1.76 or higher	1.63 or higher	1.35 or higher	1.20 or higher	1.05 or higher	0.94 or higher
Excellent	1.34 to 1.75	1.32 to 1.62	1.12 to 1.34	1.00 to 1.19	0.90 to 1.04	0.82 to 0.93
Good	1.19 to 1.33	1.14 to 1.31	0.98 to 1.11	0.88 to 0.99	0.79 to 0.89	0.72 to 0.81
Fair	1.06 to 1.18	0.99 to 1.13	0.88 to 0.97	0.80 to 0.87	0.71 to 0.78	0.66 to 0.71
Poor	0.89 to 1.05	0.88 to 0.98	0.78 to 0.87	0.72 to 0.79	0.63 to 0.70	0.57 to 0.65
Very poor	0.88 or lower	0.87 or lower	0.77 or lower	0.71 or lower	0.62 or lower	0.56 or lower
	Age					
	20 or younger	20 to 29	30 to 39	40 to 49	50 to 59	60+
Females						
Superior	0.88 or higher	1.01 or higher	0.82 or higher	0.77 or higher	0.68 or higher	0.72 or higher
Excellent	0.77 to 0.87	0.80 to 1.00	0.70 to 0.81	0.62 to 0.76	0.55 to 0.67	0.54 to 0.71
Good	0.65 to 0.76	0.70 to 0.79	0.60 to 0.69	0.54 to 0.61	0.48 to 0.54	0.47 to 0.53
Fair	0.58 to 0.64	0.59 to 0.69	0.53 to 0.59	0.50 to 0.53	0.44 to 0.47	0.43 to 0.46
Poor	0.53 to 0.57	0.51 to 0.58	0.47 to 0.52	0.43 to 0.49	0.39 to 0.43	0.38 to 0.42
Very poor	0.52 or lower	0.50 or lower	0.46 or lower	0.42 or lower	0.38 or lower	0.37 or lower

*Bench press weight ratio = weight lifted divided by body weight.
Data provided by The Cooper Institute. Physical Fitness Assessments and Norms for Adults and Law Enforcement (2013). Used with permission.

TABLE 6.2 Interpretation of Lower Body Strength for Males and Females*

Males	Age				
	20 to 29	30 to 39	40 to 49	50 to 59	60+
Well above average	2.27 or higher	2.07 or higher	1.92 or higher	1.80 or higher	1.73 or higher
Above average	2.05 to 2.26	1.85 to 2.06	1.74 to 1.91	1.64 to 1.79	1.56 to 1.72
Average	1.91 to 2.04	1.71 to 1.84	1.62 to 1.73	1.52 to 1.63	1.43 to 1.55
Below average	1.74 to 1.90	1.59 to 1.70	1.51 to 1.61	1.39 to 1.51	1.30 to 1.42
Well below average	1.73 or lower	1.58 or lower	1.50 or lower	1.38 or lower	1.29 or lower

Females	Age				
	20 to 29	30 to 39	40 to 49	50 to 59	60+
Well above average	1.82 or higher	1.61 or higher	1.48 or higher	1.37 or higher	1.32 or higher
Above average	1.58 to 1.81	1.39 to 1.60	1.29 to 1.47	1.17 to 1.36	1.13 to 1.31
Average	1.44 to 1.57	1.27 to 1.38	1.18 to 1.28	1.05 to 1.16	0.99 to 1.12
Below average	1.27 to 1.43	1.15 to 1.26	1.08 to 1.17	0.95 to 1.04	0.88 to 0.98
Well below average	1.26 or lower	1.14 or lower	1.07 or lower	0.94 or lower	0.87 or lower

*Leg press weight ratio = weight lifted divided by body weight.
Data provided by The Cooper Institute, 1994. Used with permission. Study population for the data set was predominantly white and college educated. A Universal DVR machine was used to measure the 1RM.

Assessing Muscular Endurance

As with muscular strength, various assessments can provide insight into one’s muscular endurance status. This section describes the push-up test as well as some age-specific assessments for children and older adults.

Push-Up Test for Adults

The push-up test is commonly used to measure muscular endurance, which is the ability of a muscle or muscle group to exert force repeatedly over time. Many activities of daily life such as carrying groceries and household chores require repeated or sustained muscular actions. Like muscular strength, muscular endurance can be different in upper body and lower body muscles.

The goal of the push-up test is to perform as many push-ups as possible with proper form. Note that there are two different ways to perform this test for adults, one for males and one for females. For males the toes are the rear pivot point (see figure 6.1), but for females the knees are in contact with the ground (see figure 6.2). For both males and females, proper form includes keeping the back straight while pushing up to a straight-arm position and then lowering the body until the chin touches the floor. It is important to perform the push-up test as shown so you can accurately assess your performance using table 6.3.



FIGURE 6.1 Push-up for males.



FIGURE 6.2 Push-up for females.

TABLE 6.3 Push-Up Test Norms for Adults

Males	Age				
	20 to 29	30 to 39	40 to 49	50 to 59	60 to 69
Excellent	36 or more	30 or more	25 or more	21 or more	18 or more
Very good	29 to 35	22 to 29	17 to 24	13 to 20	11 to 17
Good	22 to 28	17 to 21	13 to 16	10 to 12	8 to 10
Fair	17 to 21	12 to 16	10 to 12	7 to 9	5 to 7
Needs improvement	16 or fewer	11 or fewer	9 or fewer	6 or fewer	4 or fewer
Females	Age				
	20 to 29	30 to 39	40 to 49	50 to 59	60 to 69
Excellent	30 or more	27 or more	24 or more	21 or more	17 or more
Very good	21 to 29	20 to 26	15 to 23	11 to 20	12 to 16
Good	15 to 20	13 to 19	11 to 14	7 to 10	5 to 11
Fair	10 to 14	8 to 12	5 to 10	2 to 6	2 to 4
Needs improvement	9 or fewer	7 or fewer	4 or fewer	1 or none	1 or none

Adapted by permission from Canadian Society for Exercise Physiology, 2013.

Curl-Up and Push-Up Tests for Youth

FitnessGram includes healthy fitness zone ranges for curl-ups and push-ups (4, 5). For the curl-up test, the two pieces of tape used to help guide the extent of the curl-up are placed 3 inches (7.6 cm) apart for 5- to 9-year-olds and 4.5 inches (11.4 cm) apart for 10- to 19-year-olds. Heels must stay in contact with the mat, and no pauses or rest periods are allowed (see figure 6.3 for an example of a youth performing a curl-up). Movement should be controlled (about one curl every 3 seconds or a total of 20 per minute). If the heels come up, if the fingers do not touch the far tape, or if the child is unable to maintain a continuous cadence, the test is over and the final count should be recorded (a total of 75 curl-ups is considered maximal). Healthy ranges are shown in table 6.4.

For the push-up test, the hands are placed slightly wider than the shoulders and the legs are out straight (see figure 6.4). The back should remain in a straight line from head to toes throughout the test. The body is lowered until the elbows are at a 90-degree angle and the upper arms are parallel with the floor. Then, arms should be straightened fully to return to the starting position. The test is continued as long as these form requirements are met and the movement is continuous (no rest stops are allowed). Record the maximal number completed. Boys and girls follow the same protocol. Healthy ranges are found in table 6.4.

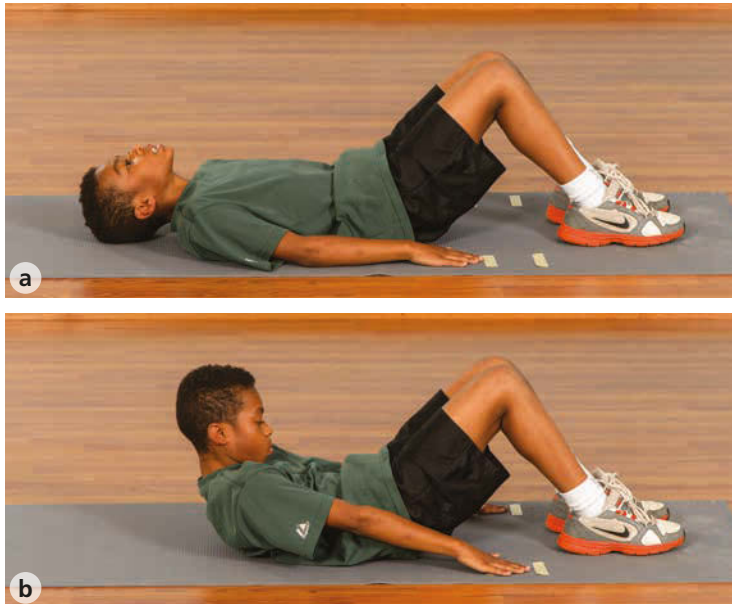


FIGURE 6.3 Curl-up for youth.

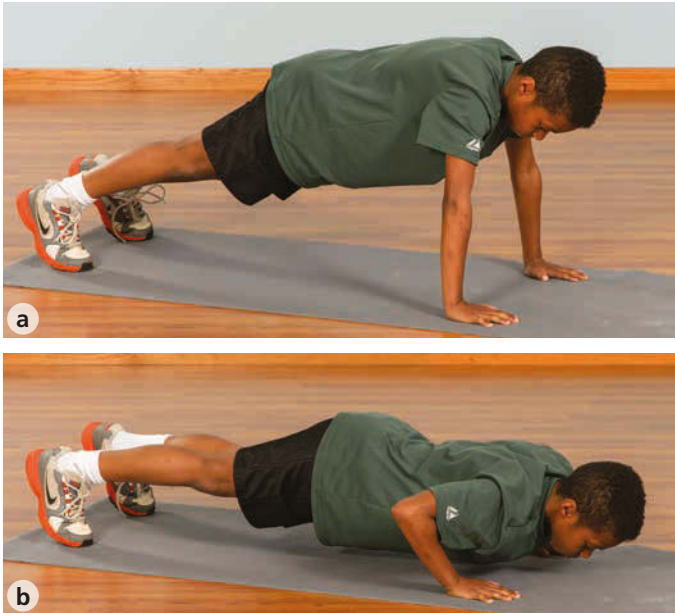


FIGURE 6.4 Push-up for youth.

TABLE 6.4 FitnessGram Standards for Healthy Fitness Zone* for Curl-Ups and Push-Ups for Youth

Age	Curl-up		Push-up	
	Boys	Girls	Boys	Girls
5	2 or more	2 or more	3 or more	3 or more
6	2 or more	2 or more	3 or more	3 or more
7	4 or more	4 or more	4 or more	4 or more
8	6 or more	6 or more	5 or more	5 or more
9	9 or more	9 or more	6 or more	6 or more
10	12 or more	12 or more	7 or more	7 or more
11	15 or more	15 or more	8 or more	7 or more
12	18 or more	18 or more	10 or more	7 or more
13	21 or more	18 or more	12 or more	7 or more
14	24 or more	18 or more	14 or more	7 or more
15	24 or more	18 or more	16 or more	7 or more
16	24 or more	18 or more	18 or more	7 or more
17	24 or more	18 or more	18 or more	7 or more
17+	24 or more	18 or more	18 or more	7 or more

*The values listed represent the healthy fitness zone and indicate that the child has a sufficient fitness level to provide important health benefits. Being below the value listed indicates a need for improvement. Adapted by permission from The Cooper Institute, 2017, pp. 86, 87.

Chair Stand Test for Older Adults

The chair stand test is used to assess lower body strength in older adults, which is important in daily activities such as climbing stairs; walking; and getting out of a chair, bathtub, or car. For the chair stand test, fold your arms across your chest and count the number of times that you can stand from a seated position in 30 seconds (see figure 6.5) (24). Normal ranges are shown in table 6.5; if your score is over the range listed, consider yourself above average and if your score falls short of the range listed, consider yourself below average.



FIGURE 6.5 Chair stand for older adults.

Arm Curl Test for Older Adults

An arm curl test is used to assess upper body strength in older adults, which is important for daily activities such as carrying groceries or small children. The arm curl assessment, as shown in figure 6.6, is used to determine upper body muscular fitness. This test involves counting the number of dumbbell curls you can complete in 30 seconds. Men should use an 8-pound (3.6 kg) dumbbell, and women should use a 5-pound (2.3 kg) dumbbell. Normal ranges are shown in table 6.5; if your score is over the range listed, consider yourself above average and if your score falls short of the range listed, consider yourself below average.



FIGURE 6.6 Arm curl for older adults.

TABLE 6.5 Normal Ranges for Fitness Test Scores for Older Adults

Males	Age						
	60–64	65–69	70–74	75–79	80–84	85–89	90–94
Chair stand test (number of stands)	14–19	12–18	12–17	11–17	10–15	8–14	7–12
Arm curl test (number of repetitions)	16–22	15–21	14–21	13–19	13–19	11–17	10–14
Females	Age						
	60–64	65–69	70–74	75–79	80–84	85–89	90–94
Chair stand test (number of stands)	12–17	11–16	10–15	10–15	9–14	8–13	4–11
Arm curl test (number of repetitions)	13–19	12–18	12–17	11–17	10–16	10–15	8–13

Adapted by permission from R.E. Rikli and C.J. Jones, 2013, pp. 89, 90.

Muscular fitness assessments that are consistent with each individual's training experience and fitness goals can provide useful information. In addition to comparing performance to that of others of the same age and sex, periodic assessments can help to gauge the effectiveness of your resistance training program. For safety purposes, individuals with health concerns should seek consultation from a health care provider before performing any fitness test.

Fundamental Principles of Resistance Training

Improvements in muscular fitness occur only if the resistance training program is based on sound training principles and is prudently progressed over time (7, 21). Although factors such as your initial level of fitness, genetics, nutrition, and motivation will influence the rate and magnitude of adaptation that occurs, you can maximize the effectiveness of your resistance training by addressing three fundamental principles: progressive overload, regularity, and specificity.

Progressive Overload

The progressive overload principle states that to enhance muscular fitness, you must exercise at a level beyond the point to which your muscles are accustomed. This goes back to the idea of having to stress the muscle to get a positive response. Doing the same workout month after month will not maximize benefits. The principle of progression refers to consistently boosting the training stimulus or load at a rate that is compatible with the training-induced adaptations that are occurring (21). Following the principle of progressive overload requires that you provide your muscles with a new stimulus when they have adapted to the current overload. You can do this in a variety of ways:

- *Increase the number of repetitions.* Typically, 8 to 12 repetitions is recommended for muscular fitness (for middle-age and older adults starting exercise, 10 to 15 repetitions is recommended). People focusing on strength development may select fewer repetitions, whereas those focusing on muscular endurance may include up to 15 to 20 repetitions (1).
- *Increase the number of sets for a given muscle group.* You could do additional sets of the same exercise, or you could add another exercise that targets the same muscle group. For example, the chest muscles could be trained with two sets of the chest press or one set of the chest press and one set of the dumbbell fly.
- *Increase the resistance.* The increase in weight needed will vary depending on the exercise but is often prescribed according to the increments available (e.g., next-weight dumbbell, increasing by one plate on a weight machine).

When providing an overload, select one of these options at a time. Although you want to provide a new stress on the muscle, you do not want to overtax the muscle or supporting structures to the point of injury.

Although every training session does not have to be more intense than the last session, the principle of progressive overload states that the training program needs to be increased gradually over time to realize gains. For example, if you have been able to easily complete a given workout for a couple of exercise sessions, it may be time to make changes to provide an overload once again in order to keep the resistance training program fresh, challenging, and effective. If you are able to perform a given exercise for one or two repetitions over your target number for two training sessions

in a row, this indicates that you are ready to increase the resistance while returning to the original target repetition range.

Regularity

The principle of regularity states that exercise must be performed several times per week on a habitual basis to enhance physical fitness. Although training once per week may maintain training-induced gains, more frequent workouts are needed to optimize gains in health and fitness (1). In short, the adage “use it or lose it” is true because you will lose strength gains if you do not progress your program over time and perform resistance training on a regular basis (21). Although consecutive days of heavy strength training for the same muscle groups are not recommended, regularly training each major muscle group two to three times per week, with at least 48 hours separating training sessions for the same muscle group, is recommended to enhance muscular fitness.

Specificity

The principle of specificity refers to the distinct adaptations that take place as a result of the training program. In essence, every muscle or muscle group must be trained to make gains in muscular fitness (see figure 6.7 for the location of the major muscle groups in the body). Exercises such as the squat and leg press can be used to enhance lower body strength, but these exercises will not affect upper body strength. What's more, the adaptations that take place in a given muscle or muscle group will be as simple or as complex as the stress placed on them. For example, because tennis requires multi-joint and multidirectional movements, it seems prudent for tennis players to perform resistance exercises that mimic the movements of the sport. For tennis players who need strong leg muscles to move across the court, lunges are unbeatable exercises to improve lower body performance. Lunges performed in different directions actually simulate steps used in game situations.

The specificity principle can also be applied to the design of resistance training programs for adults who want to enhance their abilities to perform activities of daily life such as stair climbing and household chores, which also require multijoint and multidirectional movements. For example, climbing stairs may be difficult as a result of poor lower body strength. By sensibly progressing from single-joint exercises such as leg extensions to multijoint exercises such as leg presses and dumbbell step-ups, you can improve your stair-climbing ability. These multijoint exercises specifically strengthen the quadriceps and gluteals, which are used in stair climbing.

Resistance Training Workout Components

The general format of an aerobic training session (as described in chapter 5) can be applied to resistance training as well. Before beginning a session, you should perform a warm-up to prepare your muscles for the conditioning phase of the workout. The conditioning phase is the main focus, and you should follow it with a cool-down.

Warm-Up

The warm-up for resistance training should include 5 to 10 minutes of low- to moderate-intensity aerobic activities and muscular endurance activities (lower resistance with a

Resistance Training Guidelines for Healthy Adults

Following are a few guidelines for resistance training if you are a healthy adult (1):

- Select a weight that allows you to perform 8 to 12 repetitions per set (10 to 15 repetitions for middle-age and older adults who are starting exercise).
- Train each major muscle group for a total of two to four sets (beginners can benefit from one set, which may reduce soreness and enhance adherence).
- Perform each set to the point of muscle fatigue but not failure.
- Rest for 2 to 3 minutes between sets to improve muscular fitness.
- Perform 8 to 10 exercises with proper technique.
- Resistance train two to three days per week on alternate days (48 hours is recommended between sessions to allow the muscles to recover).
- Continually progress the training program to optimize long-term adaptations.

higher number of repetitions, such as 10 to 15 repetitions). These activities will increase your body temperature and prepare your body for the demands of the resistance training workout (1).

Muscle Conditioning Phase

Despite various claims about what constitutes the best resistance training program, it does not appear that one optimal combination of sets, repetitions, and exercises promotes long-term adaptations in muscular fitness in everyone. Rather, you can alter many program variables to achieve desirable outcomes provided that you follow the fundamental training principles as discussed in this chapter. The program variables to consider are choice of exercise, order of exercise, training weight (which determines the number of repetitions), number of sets, repetition velocity, and rest periods between sets and exercises (see Resistance Training Guidelines for Healthy Adults).

Exercise Choice

A limitless number of exercises can be used to enhance muscular fitness. Exercises can generally be classified as single joint (i.e., body part specific) or multijoint (i.e., structural). The dumbbell biceps curl and leg extension are examples of single-joint exercises that isolate a specific body part (biceps and quadriceps, respectively), whereas the chest press and squat are multijoint exercises that involve two or more joints. Although it is important to incorporate multijoint exercises into a resistance training program, be sure to select exercises that are appropriate for your exercise technique experience and training goals. When learning any new exercise, start with a light weight to master the technique of the exercise before increasing the weight. To maximize gains and minimize the risk of injury, all resistance training exercise should be performed with proper exercise technique in a controlled manner.

Your choice of exercise should also promote muscle balance across joints and between opposing muscle groups (e.g., quadriceps and hamstrings). Of particular importance is the inclusion of exercises for the abdominal and low back musculature. It is not uncommon for beginners to focus on strengthening the chest and biceps and not spend adequate time strengthening the abdominal muscles and lower back.

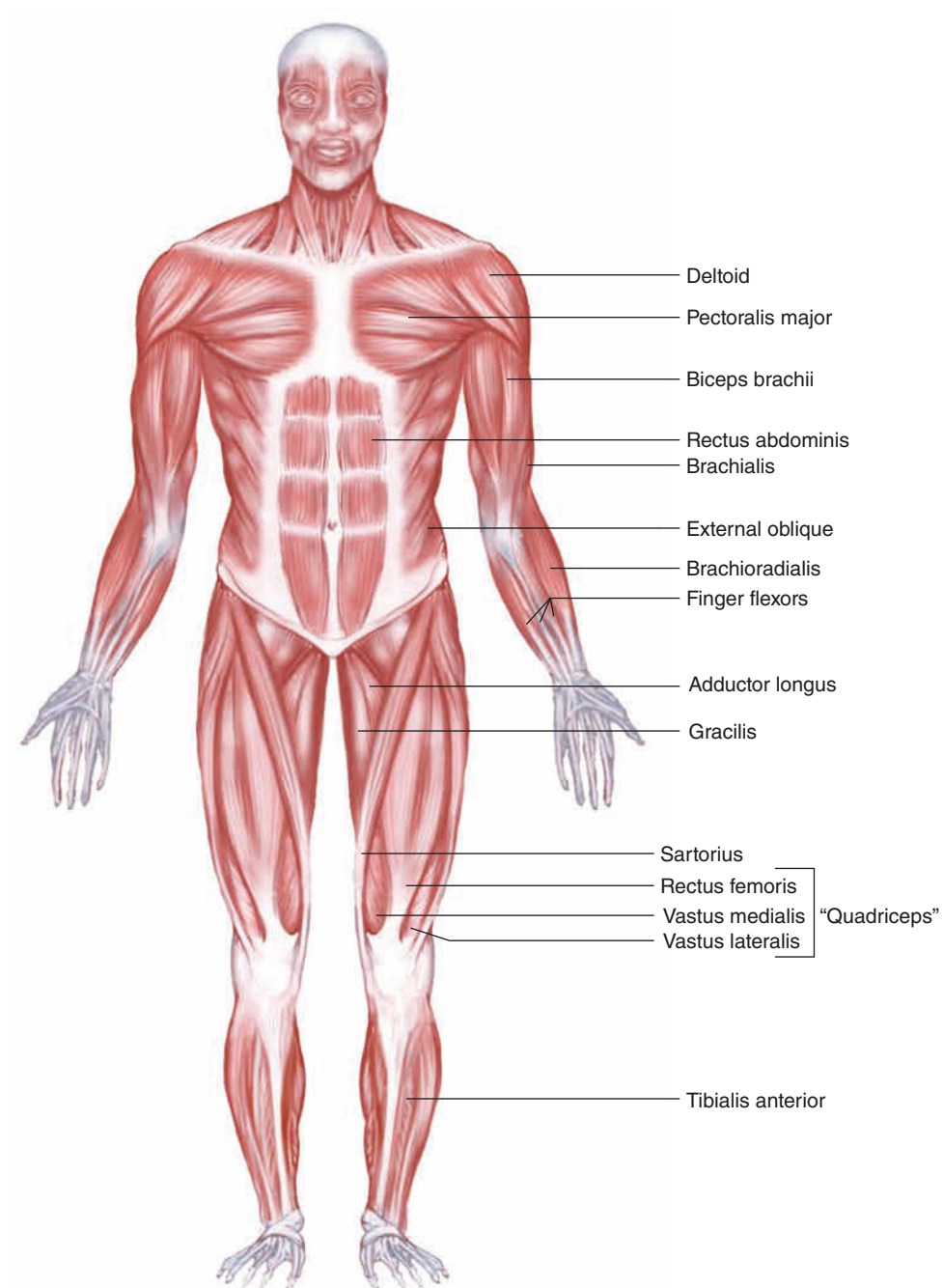


FIGURE 6.7a Major muscle groups in the body: front view.

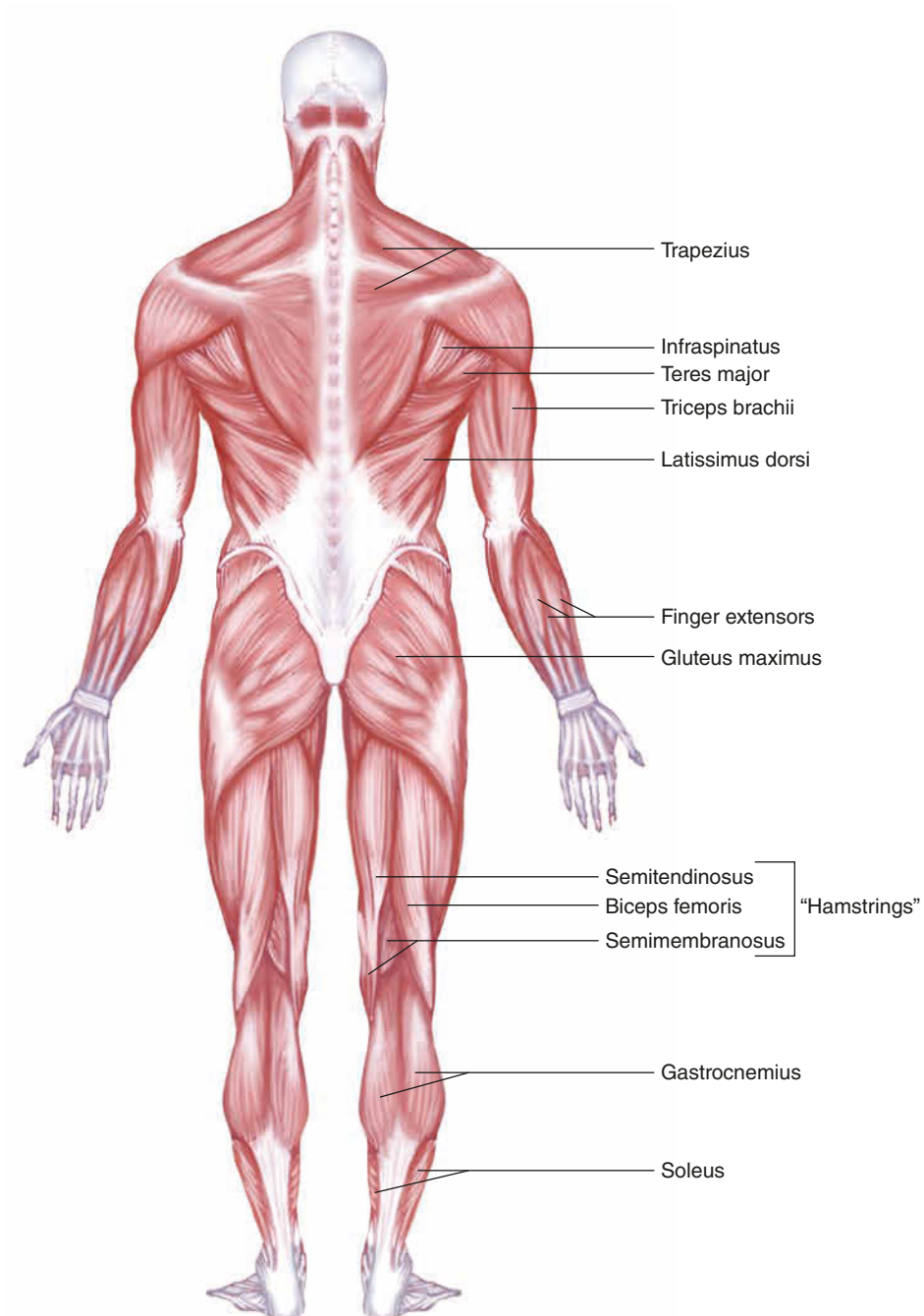


FIGURE 6.7b Major muscle groups in the body: back view.

Strengthening the midsection, or trunk area, may not only enhance body control during performance of free weight exercises such as the squat, but may also reduce the risk of injury (10). The resistance training program suggestions in this chapter promote muscle balance by including the appropriate muscle groups (see table 6.6 and figure 6.8).

Exercise Order

There are many ways to arrange the sequence of exercises in a resistance training session. Traditionally, large-muscle group exercises are performed before smaller-muscle group exercises, and multijoint exercises are performed before single-joint exercises.

TABLE 6.6 Resistance Training Workout Muscle Guide

Body area	Exercise
Hips and legs (gluteals, quadriceps, hamstrings)	Machine leg press (p. 128) Dumbbell squat (p. 129) Dumbbell step-up (p. 129) Ankle weight hip flexion and extension (p. 130) Band leg lunge (p. 131)
Legs (quadriceps)	Machine leg extension (p. 131) Ankle weight knee extension (p. 132)
Legs (hamstrings)	Machine leg curl (p. 132) Ankle weight knee flexion (p. 133)
Chest (pectoralis)	Machine chest press (p. 133) Dumbbell chest press (p. 134) Band seated chest press (p. 134) Modified push-up (p. 135) Push-up (p. 107-108)
Back (latissimus dorsi)	Machine lat pulldown (p. 135) Machine seated row (p. 136) Dumbbell one-arm row (p. 136) Band seated row (p. 137)
Shoulders (deltoid)	Machine overhead press (p. 137) Dumbbell lateral raise (p. 138) Dumbbell or band upright row (p. 138-139)
Arms (biceps)	Machine biceps curl (p. 139) Dumbbell or band biceps curl (p. 140)
Arms (triceps)	Machine triceps press (p. 141) Dumbbell lying triceps extension (p. 141) Band triceps extension (p. 142)
Low back (erector spinae)	Machine back extension (p. 142) Prone plank (p. 143) Kneeling hip extension (p. 143)
Abdominal muscles	Machine abdominal curl (p. 144) Curl-up (p. 144) Diagonal curl-up (p. 145)

Following this exercise order allows you to use heavier weights on the multijoint exercises because fatigue will be less of a factor.

Perform more challenging exercises earlier in the workout when your neuromuscular system is less fatigued. In general, it seems reasonable to follow the priority system of training in which exercises that will most likely contribute to enhanced muscular fitness are performed early in the training session. The sample resistance training programs presented in this chapter include exercises that reflect this sequence (see table 6.6 and figure 6.8).

Number of Repetitions

One of the most important variables in the design of a resistance training program is the amount of weight used for an exercise (7). Gains in muscular fitness are influenced

FIGURE 6.8

Sample resistance training programs.

Stage*	Exercises**	Number of sets	Number of repetitions	Number of days per week***
Beginner	<i>Moving through this level typically takes about two to three months, although you should remain at this level until you feel comfortable enough to advance.</i>			
	Do a total of six exercises. Select one exercise from each of the following body areas: hips and legs, chest, back, shoulders, low back, and abdominal muscles.	1 to 2	8 to 12 (10 to 15 for older adults)	2 to 3
Intermediate to established	<i>Moving through the intermediate to the established level typically takes 3 to 12 months depending on your level of consistency.</i>			
	Do a total of 10 exercises. Select one exercise from each of the following body areas: hips and legs, quadriceps, hamstrings, chest, back, shoulders, biceps, triceps, low back, and abdominal muscles.	2	8 to 12 (10 to 15 for middle-age and older adults starting exercise)	2 to 3
More advanced (complete all 15 exercises)	Do a total of 10 exercises. Select two exercises from each of these larger-muscle group areas: hips and legs, quadriceps, hamstrings, chest, and back.	2 to 3	8 to 12	2 to 3
	Do a total of five exercises. Select one exercise from each of these smaller-muscle group and trunk areas: shoulders, biceps, triceps, low back, and abdominal muscles.	2	8 to 12	2 to 3

*The time spent at each stage will depend on your muscular fitness level. Transition slowly between the stages (e.g., over time a beginner can add additional exercises or increase the number of sets to move toward the intermediate level of resistance training).

**Different exercises can be performed on different days.

***Schedule your training days so that at least 48 hours separates training sessions that target the same muscle group.

by the amount of weight lifted, which is inversely related to the number of repetitions you can perform. As the weight increases, the number of repetitions you can perform decreases. Although you should never sacrifice proper form, the training weight should be challenging enough to result in at least a modest degree of muscle fatigue during the last few repetitions of a set. If this does not occur, you will not achieve the desired gains from your resistance training program.

Because heavy weights are not required to increase the muscular strength of beginners, weights corresponding to about 60 to 80 percent of the 1RM for 8 to 12 repetitions are recommended for adults (10 to 15 repetitions for middle-age and older adults with limited resistance training experience) (1). Although weights that can be lifted more than 15 times are effective for increasing local muscular endurance, light weights rarely result in meaningful gains in muscular strength. If you are a beginner, the best approach is to first establish a target repetition range (e.g., 8 to 12), and then by trial and error determine the maximum load you can handle for the prescribed number of repetitions. If multiple sets of an exercise are performed, the first set may be performed for 12 repetitions before fatigue occurs whereas the last set may be performed for about 8 repetitions.

Although it may take two to three workouts to find your desired training weight on all exercises, keep in mind that the magnitude of your effort will determine the outcome of your strength training program. For example, training within an 8RM to 12RM zone means that you should be able to perform no more than 12 repetitions with a given weight using proper exercise technique. Simply performing an exercise for 8, 9, 10, 11, or 12 repetitions does not necessarily mean you are training within the 8RM to 12RM zone. You should be stopping because of the onset of muscle fatigue, not just because you have reached a predetermined number. However, regardless of the number of repetitions, it is important to maintain proper technique on every repetition to optimize adaptations and reduce the risk of injury.

Number of Sets

The number of sets performed in a workout is directly related to the overall training volume, which reflects the amount of time the muscles are being exercised. For beginners, even one set can provide benefits. Healthy adults should perform two to four sets for each muscle group to achieve muscular fitness goals (1). Although single-set protocols can enhance your muscular strength if you are a beginner, multiple-set pro-

Q&A

When should the weight lifted be increased?

Consider how many repetitions are currently possible. For example, initially it may be possible to lift a 20-pound (9 kg) barbell only eight times. As training continues and muscular fitness improves, this repetition number increases from 8 to 12 before fatigue (i.e., repetitions number 11 and 12 are a bit of a struggle to complete). Increasing the repetitions is one way to overload the muscle. When you are able to easily complete 12 repetitions in two consecutive training sessions, this is evidence that the muscles have adapted to the overload and now it is time to progress to a higher weight to provide greater resistance. The repetition number will drop back and the process of increasing number of repetitions from 8 to 12 will start over again.

protocols have proven more effective in the long term, with evidence of a dose response for the number of sets per exercise (14, 19). That is, greater gains in muscular fitness can be expected with additional sets per exercise (up to a point). What's more, you do not need to perform every exercise for the same number of sets. As a general recommendation, perform more sets of large-muscle group exercises than of smaller-muscle group exercises.

You can use different combinations of sets and exercises to vary the training stimulus, which is vital for long-term gains. For example, if you complete one set of two different exercises for the same muscle group (e.g., leg press and leg extension), the quadriceps on the front of the thigh will have performed two sets. From a practical standpoint, your health status, fitness goals, and time demands should determine the number of sets you perform per muscle group.

Repetition Velocity

Strength-building exercises should be performed at a controlled, or moderate, velocity during the lifting and lowering phases. Movement control can be defined as the ability to stop any lifting or lowering action at will without momentum carrying the movement to completion. Uncontrolled, jerky movements not only are ineffective but also may result in injury. Intentionally slow velocities with a relatively light weight (e.g., a 5-second lifting phase and a 5-second lowering phase) may be useful to enhance muscular endurance, but this type of training is not recommended to optimize gains in muscular strength (20). Although different movement speeds have proven effective, if you are a beginner, you should perform each repetition at a moderate speed, with about 2 seconds for the lifting phase and 3 seconds for the lowering phase. A longer lowering phase places more emphasis on the eccentric muscle action, which is important for muscle growth and strength development (20).

Rest Periods Between Sets and Exercises

The length of the rest period between sets and exercises is an important but often overlooked training variable. In general, the length of the rest period influences energy recovery and training adaptation. For example, if your primary goal is muscular strength, heavier weights and longer rest periods of 2 to 3 minutes are needed, whereas if your goal is muscular endurance, lighter weights, higher repetitions, and shorter rest periods of 30 to 60 seconds are required (20). Obviously, the heavier the weight is, the longer the rest period should be if the training goal is to maximize strength gains.

Cool-Down

The cool-down brings the body systems back to resting levels. Just as the warm-up led into the conditioning phase, the cool-down helps to transition the body from the higher demands of the conditioning phase to the lower levels of physiological demand seen at rest. Shifting to moderate-intensity and then low-intensity aerobic and muscular endurance activity will lower your heart rate and blood pressure gradually and safely (1). See Safety First for additional ways to maximize safety when training.

Types of Resistance Training

Provided that you adhere to the fundamental principles of training, you can use almost any type of resistance training to enhance muscular fitness. Some equipment is relatively

Safety First

Your resistance training program should be based on your health status, fitness training experience, and goals. As discussed in chapter 2, you should assess your health status before participating in strength-building activities. In some cases, specialized exercise programs are needed for those with preexisting medical conditions such as high blood pressure, heart disease, or diabetes. Thus, if you have a medical concern or issue, you should consult with your health care provider before resistance training.

Recognizing that resistance training to improve general fitness is different from training to enhance sport performance will further promote the development of and adherence to safe, effective, and enjoyable programs. If you have little experience with resistance training, you are strongly encouraged to seek instruction from a qualified fitness professional, because most injuries are the result of improper exercise technique or excessive loading (13, 17). Qualified fitness professionals can provide instruction on proper warm-up procedures, offer advice on specific methods of progression, and monitor the magnitude of your effort, which in turn can have a positive impact on training adaptations (16, 22).

Knowing proper breathing techniques will help you avoid the Valsalva maneuver, which can occur if you hold your breath while lifting. Not exhaling can increase pressure in the chest cavity, which can increase blood pressure to harmful levels. To avoid this effect, continue to breathe normally by inhaling before you start the lift, exhaling during the lifting–exertion phase (as you lift against gravity), and then inhaling again as you return to the starting position. Using this technique will allow you to lift weight correctly and safely.

Following are general safety recommendations for designing and performing a resistance training program:

- *Maintain a regular breathing pattern when lifting and lowering weights.* Do not hold your breath; rather, inhale before you start the lift, exhale during the lift, and inhale as you return to the starting position.
- *Make sure the exercise environment is well lit, clean, and free of clutter.* Tripping or falling over resistance training equipment can be avoided by following this guideline.
- *Learn proper exercise technique from a qualified fitness professional.* If you have little

easy to use; other equipment requires balance, coordination, and high levels of skill. A decision to use a certain type of resistance training should be based on your health status, fitness goals, training experience, and access to professional fitness instruction if needed. Common types of resistance training involve the use of weight machines; free weights; body weight exercises; and a broadly defined category that involves the use of balls, bands, and elastic tubing (table 6.7 summarizes the advantages and disadvantages of various types of resistance training). These types of resistance training typically include dynamic movements that involve a lifting (concentric) and lowering (eccentric) phase through a predetermined range of motion.

Weight machines train all the major muscle groups and can be found in most fitness centers. They are relatively easy to use because the exercise motion is controlled by the machine. For this reason, weight machines are a good option if you have not done resistance training before, are relatively new to this type of training, or are out of shape or deconditioned. Also, weight machines are ideal for isolating muscle groups. As a result, they often do not mimic sport activities or activities of daily life as well as some free weight exercises do. For general health and convenience, however, they

experience with resistance training, have someone with appropriate qualifications show you how to do resistance training exercises and assist you with making any needed adjustments.

- *Perform warm-up and cool-down activities.* Taking time for warming up and cooling down helps your body to transition safely into and out of your workout.
- *Move carefully around the strength training area.* Resistance training by its nature is equipment intensive. Dumbbells, barbells, and weight plates are all potential tripping hazards.
- *Do not use broken or malfunctioning equipment.* Check for frayed belts or cables before using any resistance training machine. Fittings should be tight, and all belts and cables should be in good condition.
- *Use collars on all plate-loaded barbells and dumbbells.* Collars are devices placed on the ends of barbells and dumbbells to hold the individual weight plates in position. Without these fasteners in place, the weight plates could shift or even fall off, causing injury.
- *Be aware of proper spotting procedures when using free weights.* A spotter is a person who is in a position to assist you when you are using free weights. Because free weights are not supported by cables or any other devices, a spotter's role is to help guide or lift a weight if you have difficulty with the resistance.
- *Avoid jerky, uncontrolled movements while resistance training.* Maintaining controlled movements maximizes the benefits of your workout and also helps you avoid injury.
- *Periodically check all training equipment.* Checking equipment for cleanliness as well as any signs of wear and tear (e.g., frayed cables or belts) and making needed corrections will help keep your resistance training sessions safe and enjoyable.
- *Regularly clean equipment pads that come in contact with the skin.* Pads become soiled with sweat; maintaining a routine of wiping off contacted surfaces promotes good hygiene.

TABLE 6.7 **Comparison of Various Types of Resistance Training**

Type	Cost	Portability	Ease of use	Muscle isolation	Functionality	Exercise variety	Space requirement
Weight machines	High	Limited	Excellent	Excellent	Limited	Limited	High
Free weights	Low	Variable	Variable	Variable	Excellent	Excellent	Variable
Body weight	None	Excellent	Variable	Variable	Excellent	Excellent	Low
Balls and cords or bands*	Very low	Excellent	Variable	Variable	Excellent	Excellent	Low

*Medicine balls, stability balls, and elastic cords or bands.

Q&A

What would be a good circuit of exercises at a fitness center?

Depending on available equipment, you will want to select exercises that target the major muscle groups. The following is an example of a program for an established exerciser with 10 body areas targeted:

- *Hips and legs*: Machine leg press or dumbbell squat
 - *Quadriceps*: Machine leg extension
 - *Hamstrings*: Machine leg curl
 - *Chest*: Machine or dumbbell chest press
 - *Back*: Machine lat pull-down or machine seated row
 - *Shoulders*: Machine overhead press or dumbbell lateral raise
 - *Biceps*: Machine biceps curl or dumbbell biceps curl
 - *Triceps*: Machine triceps press or dumbbell lying triceps extension
 - *Low back*: Machine back extension
 - *Abdominal muscles*: Machine abdominal curl
-

provide an effective method of resistance training. Although weight machines fit the typical male or female, smaller body size may require a seat pad or back pad to adjust body position to create a better fit.

Free weights, such as barbells and dumbbells, are inexpensive and can be used for a wide variety of exercises that require greater balance and coordination. Although it may take longer to master proper exercise technique using free weights compared to weight machines, proper fit is not an issue because one size fits all. Free weights also offer a greater variety of exercises than weight machines because they can be moved in many directions. Another benefit of free weights is that they require the use of additional stabilizing and assisting muscles to hold the correct body position to perform an exercise correctly. As such, free weight training can occur in different planes of motion and is ideal for enhancing performance during activities of daily life. This is particularly true regarding the use of dumbbells because they train each side of the body independently. However, unlike weight machines, several free weight exercises, such as the bench press, require the aid of a spotter who can assist the lifter in case of a failed repetition. Spotters should be able to handle the weight lifted and should know when to intervene.

Body weight exercises such as push-ups, pull-ups, and curl-ups are among the oldest modes of strength training. Obviously, a major advantage of body weight training is that equipment is not needed and a variety of exercises can be performed. On the other hand, a limitation of body weight training is the difficulty in adjusting the body weight to the strength level of the person. Exercise machines that allow you to perform body weight exercises such as pull-ups and dips using a predetermined percentage of your body weight are available. Even if you do not have the strength to lift your entire body weight, these machines provide assistance, allowing participants of all abilities to incorporate body weight exercises into their strength training programs and feel good about their accomplishments.

Stability balls, medicine balls, and elastic tubing are inexpensive, safe, and effective alternatives to weight machines and free weights. Stability balls are lightweight, inflatable balls (about 45 to 75 centimeters in diameter) that add the elements of balance and coordination to any exercise. Medicine balls come in a variety of shapes and sizes (about 2 pounds to over 20 pounds, or 1 kilogram to over 9 kilograms) and stress muscles as you hold, catch, and throw them. Training with elastic rubber cords, or bands, involves generating force to stretch the cord and then returning the cord in a controlled manner to its unstretched state. The more the cord is stretched, the greater the force needed to move through the range of motion. Different colors of cords reflect different amounts of resistance.

Q&A

Can muscular fitness be improved without access to a fitness center?

Although membership at a fitness center has many advantages, you can also improve your muscular fitness at home. Resistance bands and ankle weights are relatively inexpensive purchases. In addition to exercises that use body weight, these can target the major muscle groups. For example, a beginner program with six exercises might look like this:

- *Hips and legs:* Ankle weight hip flexion and extension or band leg lunge
- *Chest:* Band seated chest press or modified push-up
- *Back:* Band seated row
- *Shoulders:* Band upright row
- *Low back:* Prone plank or kneeling hip extension
- *Abdominal muscles:* Curl-up

Your Resistance Training Program

Your resistance training program needs to take into account your current muscular fitness level. Beginner, intermediate, and more advanced sample programs are outlined in figure 6.8 (for a workout guide that groups exercises into the appropriate body areas, see table 6.6). If you have no resistance training experience or have not trained for several months or years, you should begin resistance training by following a general program in which weights are light to moderate and the focus is on learning proper exercise technique. Also, it is always a good idea for beginners to receive instruction on proper exercise technique and training guidelines from a qualified fitness professional.

Avoid the common mistake of doing too much too soon. Give your body a chance to adapt gradually to the physical stress of resistance training while making fitness gains. Use the initial weeks to increase your body's ability to tolerate the stress of resistance training gradually to minimize muscle soreness. The aim is to develop healthy habits early on so that resistance training becomes an enjoyable, meaningful, and long-lasting experience. Regardless of how much weight others can lift, go slowly during the first few weeks as you build a foundation for more advanced training programs in the future.

As indicated in the sample programs in figure 6.8, if you are a beginner, you should perform one or two sets of six exercises with a moderate weight. Of course, regardless of your level of experience, you should use lighter loads when you are learning a new

exercise or attempting to correct any flaws in your exercise technique. Also, keep in mind that you do not have to perform every exercise for the same number of sets. This preparatory period is designed to gradually enhance your physical abilities as you start the process of resistance training. If you have a very low level of fitness, you may need a longer period of time before you can participate in a resistance training program designed to maximize gains in muscular fitness. A major goal of this training phase is to learn correct form and technique for a variety of upper body, lower body, and midsection exercises while practicing proper training procedures. Table 6.6 outlines resistance training exercises that use weight machines, free weights (dumbbells), and your own body weight.

Once you are comfortable with the level of exercise at the beginner level, you are ready to move to the intermediate level. Typically, this takes around two to three months, although this time may be shorter or longer depending on your initial fitness level. The intermediate level begins once you have progressed through the beginner level, or you can start at this level if you are already engaging in some resistance training. The intermediate activities are broader in scope than the beginner activities and also increase the overall volume (increasing the number of exercises and sets). Depending on the consistency of your training, you may spend three months to a year or more at the intermediate level.

After 6 to 12 months of consistent training, you may appropriately be classified as “established.” At this point you can continue with the intermediate-level exercise format but increase the weight, or resistance, over time (recall the concept of progressive overload). Figure 6.8 includes a “more advanced” category for those looking to increase their focus beyond health-related levels of resistance training. More advanced resistance training can provide additional muscular fitness benefits and includes exercises for different body parts on separate days of the week (thus increasing the overall training volume and the time you spend training).

By varying the program variables such as the choice of exercise and number of sets, you will start to achieve specific goals in health and fitness. Although every workout does not need to be more intense than the previous one, varying your program helps to prevent boredom and training plateaus that eventually lead to a lack of adherence and dropout. As you perform additional sets, keep in mind that your effort determines your training outcomes. Thus, feelings related to exercise exertion should be an expected and welcome part of the training process. A major goal is to gain confidence in your ability to perform strength-building exercises while maximizing training adaptations.

After the first few months of resistance training, improvements in muscular fitness occur at a slower rate. People who started resistance training with great enthusiasm sometimes become disappointed when gains in muscle strength are less dramatic during the third month of training. You need to understand that a workout that was effective during the first few months may not be effective in the long term. Once your body adapts to the training program, no additional gains will take place unless the training program is altered. In short, to make continual gains in muscular fitness and achieve specific health and fitness goals, you need to work harder and engage in a more challenging training program. This is particularly important if you want to maximize gains in muscular fitness (21).

Because of the demands of training, you need to allow time for adequate recovery between workouts for a given muscle group. For example, more advanced lifters may perform a whole-body workout only twice per week or a greater number of sessions

per week using a split routine in which only certain muscle groups are selected on a given day. For example, a lifter may train the lower body on Monday and Thursday and the upper body on Tuesday and Friday. In any case, all lifters should appreciate the importance of adequate recovery between demanding resistance training workouts.

For continued gains in muscular fitness, you must sensibly alter your resistance training program over time so your body is continually challenged to adapt to the new demands (21). To clarify, every workout does not need to be harder than the previous workout; rather, a systematic progression of the exercise program is needed for long-term gains in muscular fitness. Even though beginners will improve at a faster rate than more experienced lifters, manipulating the program variables every couple of weeks will limit training plateaus and reduce the likelihood that you become bored with your training program and lose your enthusiasm for resistance training.

Although improving at the same rate over the long term is not possible, you have to place greater demands on the musculoskeletal system gradually if you want to

Resistance Training That Works for You

A total-body resistance training workout is an effective way to improve muscular fitness and physical performance. Although resistance training programs that split the body into selected muscle groups are popular, a total-body workout performed two to three days per week on nonconsecutive days is appropriate for most people. Such a program gives you time to learn proper exercise technique and develop a fitness base for more advanced training. The idea is to start with a general resistance training program and gradually make it more specific as your strength and confidence improve.

Because the ultimate goal is the adoption of muscular fitness exercises as a lifestyle choice, your resistance training program should be consistent with your current fitness status and personal goals. In addition, you need to consider the time you have available for training, the equipment you can access, and your strength training experience. Consider the following questions before beginning a resistance training program:

- Do you have health concerns that may limit your participation in a resistance training program?
- Do you currently participate in an exercise program?
- How much resistance training experience do you have?
- What type of resistance training equipment is available at home or at your gym?
- How much time do you have for resistance training during the week?
- What are your specific training goals?
- Would individualized instruction from a qualified fitness profession be beneficial?

Once you have answered these questions, you are ready to design a safe, effective, and enjoyable resistance training program that is consistent with your goals. This chapter provides guidance whether you are just starting or are already doing resistance training and are looking for ways to continue to improve. To make continual gains in health and fitness, you must continue, progress, and modify your program.

Using a workout card to monitor your training progress can be very helpful. On the card, record the exercises, weight lifted, and number of repetitions and sets. It is also a good idea to exercise with a training partner or fitness instructor who can serve as a spotter on selected exercises and provide assistance when needed.

make steady gains in muscular fitness. In addition to increasing the amount of weight you lift, you can also progress your training program in other ways. You can perform additional repetitions with the current weight, add more sets to your program, and incorporate different exercises or types of equipment into the program to provide progressive overload. The key to long-term training success is to make gradual changes in the program to keep it effective, challenging, and fun.

Resistance training is an essential component of adult fitness programs and can offer observable health and fitness gains when properly performed and sensibly progressed over time. The importance of the training-induced changes from resistance training should not be underestimated because they can have a meaningful impact on your physical function and quality of life (20, 30). Although many exercise options are available, resistance training programs based on sound training principles and consistent with your needs, goals, and abilities are most likely to result in favorable adaptations. In general, perform resistance training two to three days per week (with 48 hours between sessions), do two to four sets of 8 to 12 repetitions of each exercise (10 to 15 repetitions for middle-age and older adults starting exercise), and target each of the major muscle groups.

Resistance Training Exercises

Descriptions and photos for each of the exercises in the sample programs are included here (see table 6.6 for a guide showing you which exercises work specific body areas). In general, the photos depict the two ends of the range of motion for each exercise. Be sure to control your movement to reap the full benefits from each exercise.

Machine Leg Press



Adjust the machine so your knees are bent about 90 degrees, with feet flat on the foot pads (a). Your knees and feet should be in line with your hips. Exhale and push your feet and legs forward by pushing through your heels until your knees are nearly straight (b). Do not lock your knees.

Dumbbell Squat

Choose your desired or appropriate dumbbell weights. Spread your feet about shoulder-width apart; your knees and feet should be in line with your hips (a). Bend slightly at the hips and then bend your knees until your thighs are parallel to the floor (b). Your knees should not go beyond your toes. Pause briefly; then return to the starting position. Keep your chest up throughout the movement to avoid excessive forward lean.



Dumbbell Step-Up

Choose your desired or appropriate dumbbell weights. Stand with a dumbbell in each hand facing a step (or bench). Place one foot on the step (a) and then step up with the other foot while keeping torso upright (b). Step back down and return to the starting position. Repeat with opposite leg. Begin with body weight only to learn proper form.

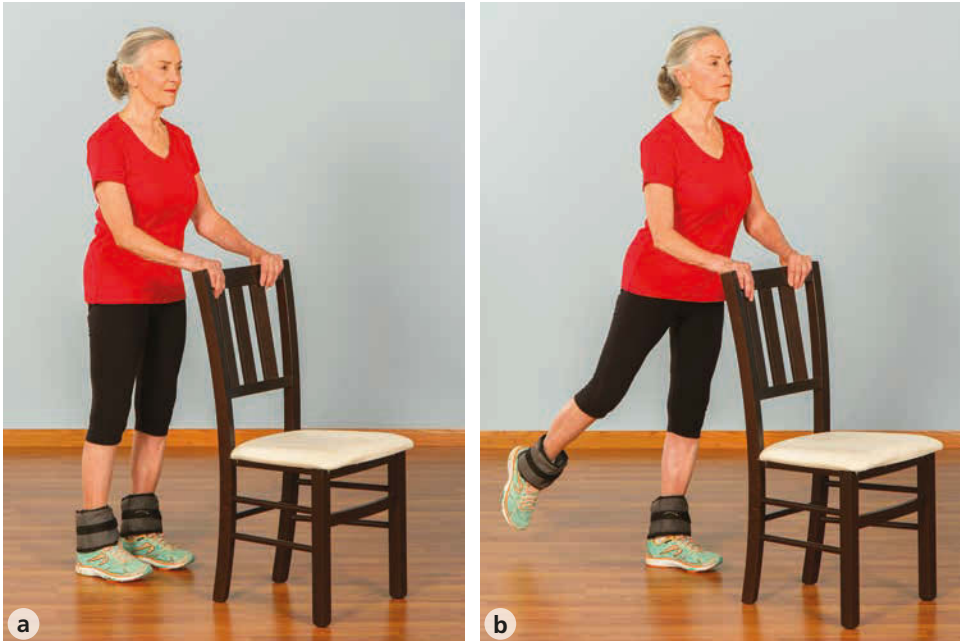


Ankle Weight Hip Flexion

Ankle weights are needed for this exercise. Stand tall with one hand on the back of a chair for balance (a). Without leaning forward, lift one knee toward your chest in a marching motion (b), pause briefly, and then return your knee to the starting position and repeat on the opposite side.



Ankle Weight Hip Extension



Ankle weights are needed for this exercise. Stand about 12 inches (30.5 cm) from a chair with your feet slightly apart. Bend forward slightly and hold on to the back of the chair for balance (a). Lift one leg backward without moving your upper body forward or bending your knee (b). Pause briefly; then return to the starting position and repeat on the opposite side.

Band Leg Lunge



Start in a stride position with one foot in the middle of the band and the other foot extended behind your body. Pull the band tight by bending your elbows to allow your hands to be at shoulder height (a). Lower your body toward the floor while keeping your shoulders over your hips and your front knee over the ankle of your front foot (b). Return to the starting position and perform the desired number of repetitions. Repeat on the opposite side.

Machine Leg Extension



Adjust the machine so your knee joints are in line with the machine's axis of rotation and the leg pads are just above your ankles (a). Straighten both knees until they are fully extended (b), pause briefly, and then return to the starting position and repeat.

Ankle Weight Knee Extension



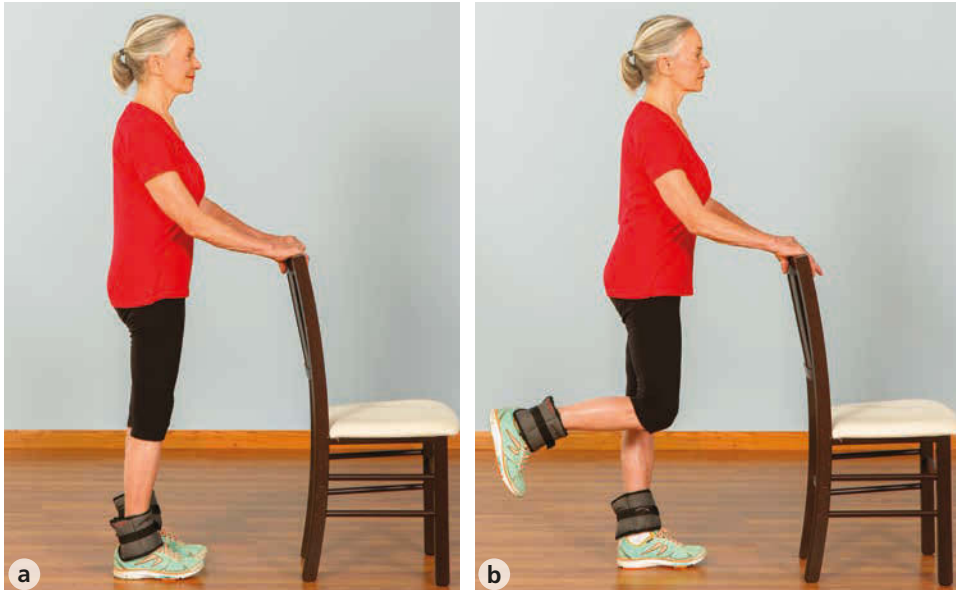
Ankle weights are needed for this exercise. Sit tall in a chair with your feet flat on the floor (a). Lift one leg by straightening your knee until the leg is parallel to the floor (b). Pause briefly; then return your leg to the starting position and repeat on the opposite side.

Machine Leg Curl



Adjust the machine so your knees are in line with the machine's axis of rotation and the roller pads are under your ankles (a). Grasp both handles. Pull the roller pad toward your hips until both knees are bent at least 90 degrees (b). Pause briefly; then return to the starting position and repeat.

Ankle Weight Knee Flexion



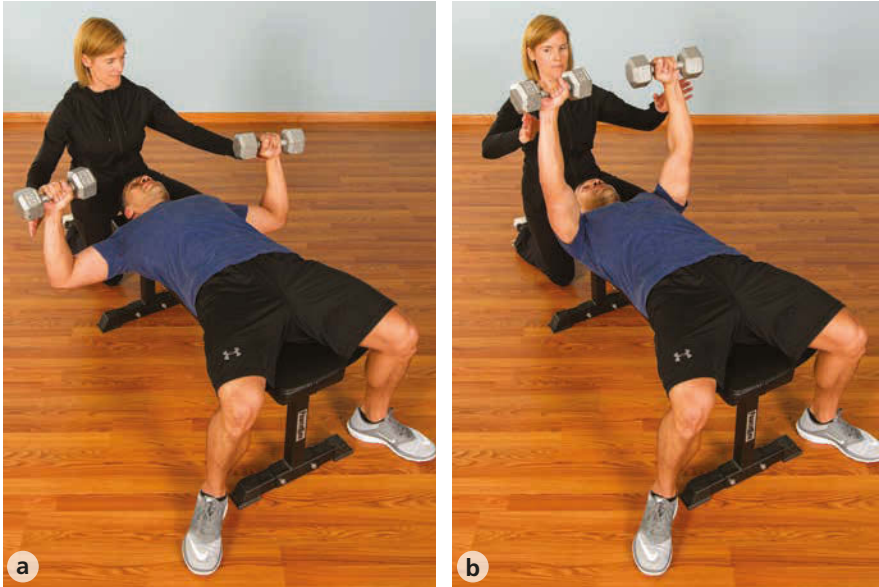
Ankle weights are needed for this exercise. While wearing ankle weights, stand tall behind a chair and grasp the chair back (a). Bend one knee and raise your foot toward your buttocks without moving your thigh (b). Pause briefly; then return to the starting position. Repeat on the other side.

Machine Chest Press



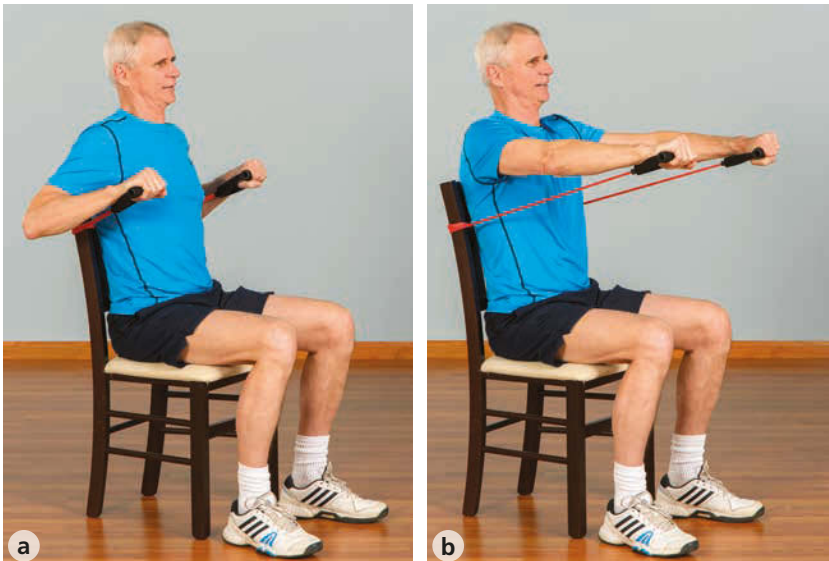
Adjust the seat so that the handles are aligned at midchest level. Sit with your back against the seat pads and grasp the bar handles with an overhand grip (a). Push the handles forward until your elbows are straight and fully extended but not locked (b). Pause briefly; then return the handles to the starting position and repeat.

Dumbbell Chest Press



Choose your desired or appropriate dumbbell weight. Lie on a bench with your knees bent and your feet flat on the floor. Your head, shoulders, back, and buttocks must maintain contact with the bench during the exercise. Hold the dumbbells at the side of your chest with your thumbs wrapped around the handles and your elbows bent about 90 degrees (a). Press the dumbbells upward over your chest until your arms are straight (b). Return to the starting position and repeat. A spotter should be nearby to assist you if needed.

Band Seated Chest Press



Choose a band color or thickness. Sit in a chair and wrap the band around the back of the chair. Hold the ends of the band at chest level with your elbows bent (a). The band tension should be tight. Press both arms straight out in front of your body (b). Pause briefly; then return to the starting position and repeat.

Modified Push-Up

Stand 2 to 3 feet (61-91 cm) from a wall and place your palms on the wall at shoulder height (a). Your palms should be placed slightly wider than your shoulders. Keeping your back straight, bend your elbows until your nose almost touches the wall (b). Pause briefly; then press away from the wall and return to the starting position. Moving your feet farther away from the wall increases the difficulty of this exercise. As you gain more strength in your upper body, progress to bent-knee push-ups on the floor (see figure 6.2) and finally to full push-ups (see figure 6.1).



Machine Lat Pull-Down



Adjust the seat height and extend your arms overhead to grasp the bar (a). Your palms should face forward with your hands slightly wider than shoulder width. Lean back slightly, and pull the bar downward to the top of your chest (b). Tuck your chin to allow the bar to freely pass in front of your face. Focus on pulling your elbows in toward your body. Return to the starting position and repeat.

Machine Seated Row



Move the seat so your shoulders are level with the machine handles and your chest is against the chest pad. Grasp the handles and sit tall with your chest up (a). Pull the handles backward while moving your shoulder blades together (b). Return to the starting position and repeat.

Dumbbell One-Arm Row



Choose your appropriate or desired dumbbell weight. Stand near the left side of the bench and place your right knee and the palm of your right hand on the bench, keeping your right arm straight and your torso almost horizontal. Hold the dumbbell in your left hand with your palm toward the bench (a). Pull the dumbbell toward the side of your chest by bending at the elbow and the shoulder (b). Return to the starting position and perform the desired number of repetitions. Repeat on the opposite side.

Band Seated Row



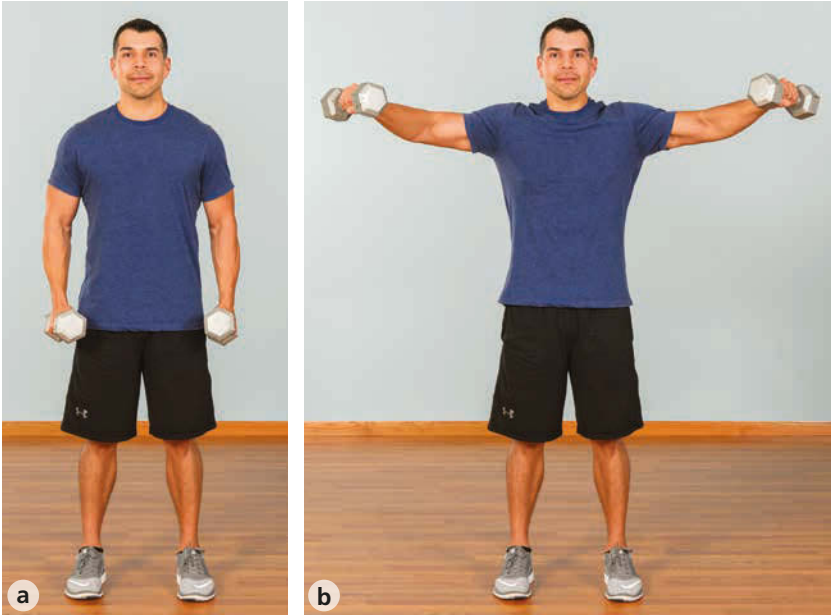
Choose a band color or thickness. Sit on the floor and wrap the band securely around both feet. The middle of the band should be placed at the center of your feet. Point your toes slightly forward to prevent the band from slipping. Fully straighten your elbows with your palms facing each other (a). The band tension should be tight in both your hands. Pull the band toward the sides of your body while keeping your back straight (b). Pause briefly; then return to the starting position and repeat.

Machine Overhead Press



Adjust the seat height so the handles are aligned with or slightly above your shoulders. Grasp the handles and sit up straight with your head, shoulders, and back against the pad and your feet flat on the floor (a). Push the weight up over your head until your arms are fully extended but not locked (b). Pause briefly and return to the starting position.

Dumbbell Lateral Raise



Choose your appropriate or desired dumbbell weight. Stand with your feet shoulder-width apart. Hold a dumbbell at the side of your body with your palms facing in and your elbows slightly bent (a). Raise both arms out to the sides until they are horizontal (b). Pause briefly; then return to the starting position and repeat.

Dumbbell Upright Row



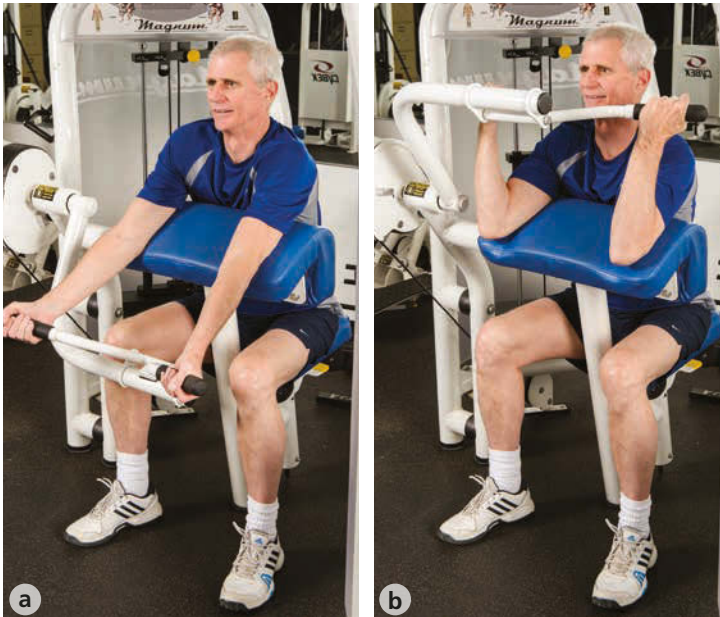
Stand tall with your feet shoulder-width apart. Hold a dumbbell in each hand with your palms facing your thighs and your elbows pointing outward (a). Bend at the elbows and lift both dumbbells to shoulder level (b). Keep your elbows pointed outward during the upward movement. Pause briefly; then lower the weights to the starting position and repeat.

Band Upright Row

Choose your appropriate or desired band. Stand with both feet placed about shoulder-width apart on top of the band. Grasp one end of the band in each hand and stand erect (a). Your palms should be facing your thighs, and the band tension should be tight. Bend at the elbows and pull the band to shoulder level (b). Keep your elbows pointed outward during the upward movement. Pause briefly; then lower your arms to the starting position and repeat.



Machine Biceps Curl



Adjust the seat height so your upper arms are resting flat against the arm pad and your elbow is aligned with the machine's axis of rotation. Grasp the handles firmly and position your body so your chest is up and your shoulders are back (a). Curl your hands toward your shoulders until your elbows are fully flexed (b). Return to the starting position and repeat.

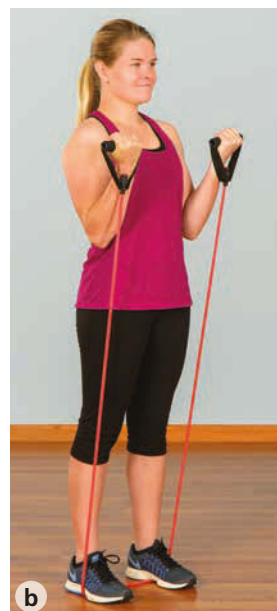
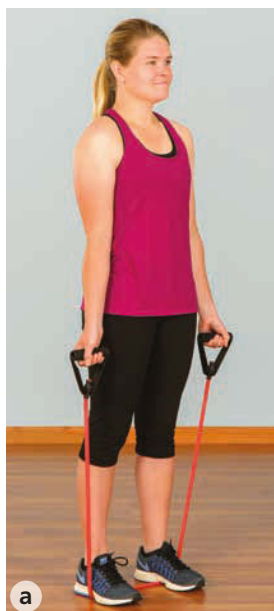
Dumbbell Biceps Curl

Choose your appropriate or desired dumbbell weights. Stand tall with your feet shoulder-width apart. Hold the dumbbells with your palms facing forward and your elbows at the sides of your body (a). Raise both dumbbells by bending your elbows until they are fully flexed (b). Keep your elbows at your sides during the entire movement. Lower the dumbbells to the starting position and repeat. This exercise can also be performed in a seated position with alternating arms.



Band Biceps Curl

Choose a band thickness or color. Stand with both feet placed about shoulder-width apart on top of the band. Grasp one end of the band in each hand and stand erect (a). Your palms should be facing forward, and the band tension should be tight. Bend your elbows until they are fully flexed (b). Keep your elbows at your sides during the entire lift. Lower your arms to the starting position and repeat. This exercise can also be performed in a seated position with alternating arms.



Machine Triceps Press

Stand tall with hands placed approximately shoulder-width apart on the bar. Grasp the bar and position your body so your chest is up and shoulders are back (a). Move the bar from the starting position until your elbows are fully extended but not locked (b). Return to the starting position and repeat.



Dumbbell Lying Triceps Extension



Choose an appropriate or desired dumbbell weight. Lie on a bench with your knees bent and your feet flat on the floor. Your head, shoulders, back, and buttocks must maintain contact with the bench during this exercise. Hold a dumbbell in each hand with your thumb wrapped around the dumbbell and both arms fully extended above your shoulders (a). Bend your elbows and slowly lower the dumbbells toward (but not touching) the side of your head (b). Return to the starting position and repeat. A spotter should be nearby to assist you if needed.

Band Triceps Extension

Choose a band thickness or color. Stand straight with your feet about shoulder-width apart. Hold one end of the band in your left hand placed near your low back and the other end in your right hand placed behind your neck (a). Move your arm up so the right elbow straightens overhead without moving your left arm (b). Pause briefly; then slowly return your right hand to the starting position and perform the desired number of repetitions. Repeat on the opposite side.



Machine Back Extension



Adjust the seat so your navel is aligned with the machine's axis of rotation. Sit with your back against the pad, your feet on the foot pad, and your arms folded across your chest (a). Slowly lean backward (extending the torso) with the back in contact with the pad (b). Pause briefly; then return to the starting position and repeat.

Prone Plank



Lie facedown on the floor with your feet behind your body. Support your weight on your knees and forearms (a). Keep your back flat and your head in line with your torso. Breathe normally as you hold the position for the desired number of seconds. To increase difficulty, lift your knees and support your weight on your toes and forearms (b).

Kneeling Hip Extension



Kneel down in the crawl position with your arms directly below your shoulders (a). Extend your right leg backward until it is parallel to the floor while keeping your shoulders and hips level (b). Pause briefly; then return to the starting position and repeat on the opposite side.

Machine Abdominal Curl



Adjust the seat so your navel is aligned with the machine's axis of rotation (a). Curl your torso forward while fully flexing your trunk (b). Pause briefly; then return to the starting position and repeat.

Curl-Up



Lie on your back with your knees bent and your feet flat on the floor (a). Place your hands on your thighs. Curl your shoulders and upper back off the floor while sliding your hands up your thighs toward your kneecaps (b). Your low back should remain in contact with the floor. Pause briefly; then return to the starting position and repeat.

Diagonal Curl-Up



Lie on your back with your knees bent and your feet flat on the floor (a). Place your hands on your thighs. Curl your shoulders and upper back off the floor while sliding your left hand toward your right kneecap (b). Your low back should remain in contact with the floor. Pause briefly; then return to the starting position and repeat on the opposite side.

Regular participation in strength-building exercises that enhance muscular fitness can offer observable health and fitness value. Muscles and bones get stronger and activities of daily life become easier to perform. A well-designed resistance training program should target all the major muscle groups and should be sensibly progressed over time to optimize adaptations and maintain interest.

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SEVEN

Increasing Your Flexibility

Flexibility may not have the same health benefits as aerobic or muscular fitness, but it is an important part of your overall physical fitness. Many activities require flexibility (e.g., golfing, swimming, dancing), and daily activities are also affected by flexibility (e.g., reaching, bending, twisting) (2, 3). Flexibility varies between individuals and is affected by joint types, differing muscle lengths, ligaments, tendons, muscles, skin, and age (17). Understanding flexibility and its role in exercise programming is essential for a well-balanced exercise regimen.

Flexibility Factors

Flexibility is the ability of a joint and surrounding muscle to move through a full or optimal range of motion (6). Improving range of motion at a joint eliminates awkward and inefficient movements, allowing you to move more fluidly. You can appreciate this throughout your day-to-day activities and in any recreation or sports you may do. Maintaining or improving your range of motion through flexibility exercises helps you move more efficiently (3, 4, 7). For example, if you improve range of motion in your hips and hamstring muscle groups, which are located at the back of your thighs, you can ease the task of reaching down to pick up a grocery bag or bending over to tie your shoes, as well as increase your stride when jogging or running.

Several factors influence flexibility, including age, sex, joint structure, and physical activity level (4, 17). Females tend to have a slightly greater range of motion at most joints than males do. This is usually explained by differences in joint structure and is often observed in joints in the upper body (e.g., shoulders, elbows, wrists, neck), with the exception of the trunk, in which males tend to have a greater range of motion than females (9). Flexibility typically decreases with age, resulting in many significant changes in the neck, shoulder, and trunk region (9, 19). You can minimize these changes by adhering to a regular stretching program. Specific activities you can incorporate into your stretching routine are outlined in this chapter.

Q&A**How can routine activities affect flexibility?**

Routine daily activities including desk work can cause slumped-forward shoulders (internally rotated humerus) as well as lower back pain. To combat this, it is important to include dynamic and static stretches of the chest, shoulders, neck, and hips.

Health and Fitness Benefits of Flexibility

Compared to less active people, active people have greater flexibility in the joints they use (14). For example, people who walk more tend to have greater flexibility in their hips and spine than people who walk less. And limited motion of a specific joint can lead to a loss in flexibility. If you spend several hours per day driving or sitting at a computer, you may find that your shoulders round forward as a result of decreased range of motion at your shoulder joints. A focus on stretching and body position can help you avoid such losses in flexibility and allow you to keep a strong upright posture.

In addition, improvements in flexibility may enhance performance in certain skills that require greater flexibility (e.g., dancing, golf) (8). However, unless you have poor flexibility at a specific joint, increasing flexibility beyond a normal range of motion does not benefit performance or decrease the risk of injury. Contrary to popular belief, there is not sufficient evidence to support the contention that preexercise stretching prevents all injuries (although there may be potential benefits related to acute injuries in repetitive activities like running [5]), or that pre- and postexercise stretching prevents muscle soreness (5, 13). However, you may experience relaxation or stress relief from participating in flexibility-focused exercises.

Assessments for Range of Motion

Several tests can be administered for flexibility assessment if one has access to some rather simple pieces of equipment such as these:

- Masking tape
- Chair
- Measuring stick (yardstick or meter stick)
- Ruler

Flexibility testing is similar to stretching exercises in that a brief warm-up should be administered before attempting the assessment. On average, 5 minutes of brisk cardiovascular activity like a brisk walk, moderate cycling, or marching in place is typically enough to increase the heart rate and provide pliability to the muscle structures.

Q&A**Does stretching prevent or reduce muscle soreness?**

Researchers have not proven that stretching before or after an exercise session adds benefits or protection against muscle soreness (11). Stretching has not been found to attenuate the structural mechanisms that contribute to soreness, including microtears to the muscle fibers, accumulation of calcium ions, cellular inflammation, or swelling.

This chapter describes a small subset of tests that are available to assess flexibility. Each joint and body segment is unique. Two assessments covered in this chapter are the sit-and-reach and the back-scratch test. These tests provide a snapshot of flexibility in the trunk and hips as well as the upper arm and shoulder.

Sit-and-Reach Assessments

The sit-and-reach is one of the most common tests of flexibility and reflects hamstring flexibility (muscles on the back of the thigh) and possible low back flexibility as well (3). There are various versions of the assessment for adults, older adults, and children.

Sit-and-Reach Test for Adults

The setup for the adult sit-and-reach test requires a yardstick and masking tape. The following steps outline the setup and measurement (10):

- A yardstick is placed on the floor and tape is placed across it at a right angle to the 15-inch (38 cm) mark. Sit with the yardstick between your legs, with legs extended at right angles to the taped line on the floor. Heels of the feet (with shoes removed) should touch the edge of the taped line and be about 10 to 12 inches (25-30 cm) apart (see figure 7.1a).
- Slowly reach forward with both hands as far as possible, holding this position approximately 2 seconds (see figure 7.1b). Be sure that you keep the hands parallel and do not lead with one hand or bounce. Fingertips can be overlapped and should be in contact with the yardstick.
- The score is the most distant point (recorded in inches) reached with the fingertips. The best of two trials should be recorded. To assist with the best attempt, you should exhale and drop your head between your arms when reaching. Make sure that your knees stay extended. Do not bend the knees as you reach forward, but also, do not press the knees down. Breathe normally during the test; you should not hold your breath at any time.
- You can track your score over time, realizing that a higher score indicates improving flexibility.



FIGURE 7.1 Sit-and-reach for adults.

Chair Sit-and-Reach Test for Older Adults

The sit-and-reach test for older adults can be done while sitting in a chair rather than on the floor (16). In addition to a sturdy chair, you will also need a ruler. Note that if you have pain when flexing, have severe osteoporosis, or have had recent knee or hip replacement, this assessment should not be attempted. The following steps outline the setup and results:

- Start in a sitting position on a chair placed securely against a wall. Move forward until you are sitting on the front edge of the chair. The crease between the top of the leg and the buttocks should be even with the edge of the chair seat.
- Keep your right leg bent and foot flat on the floor; the left leg is extended straight in front of the hip, with heel on floor and foot flexed (at approximately 90 degrees) (see figure 7.2a). With the left leg as straight as possible (but not hyperextended), slowly bend forward at the hip joint, reaching as far toward or past your toes on your left foot as possible (see figure 7.2b). Your hands should be one on top of the other with the tips of the middle fingers even. Continue breathing normally; do not hold your breath. Exhale as you reach forward and avoid bouncing or forcing the movement. Hold your reach for 2 seconds. Practice the movement twice.
- Now switch the position of your feet and keep your left leg bent and foot flat on the floor with the right leg extended straight in front of you. Repeat the test in this position.
- Whichever position results in the farthest reach is used for scoring purposes. Repeat the test two more times in this position. Using a ruler, record the number of inches short of reaching the toes (minus score) or beyond the toes (plus score). The tip of your toes at the end of your shoe represents a zero score.



FIGURE 7.2 Chair sit-and-reach for older adults.

- Use table 7.1 to interpret the results of your sit-and-reach test. Normal ranges are shown; if your score is over the range listed, consider yourself above average and if your score falls short of the range listed, consider yourself below average.

TABLE 7.1 **Normal Ranges for Chair Sit-and-Reach for Older Adults in Inches**

	Age						
	60 to 64	65 to 69	70 to 74	75 to 79	80 to 84	85 to 89	90 to 94
Males	–2.5 to +4.0	–3.0 to +3.0	–3.0 to +3.0	–4.0 to +2.0	–5.5 to +1.5	–5.5 to +0.5	–6.5 to –0.5
	Age						
	60 to 64	65 to 69	70 to 74	75 to 79	80 to 84	85 to 89	90 to 94
Females	–0.5 to +5.0	–0.5 to +4.5	–1.0 to +4.0	–1.5 to +3.5	–2.0 to +3.0	–2.5 to +2.5	–4.5 to +1.0

Adapted by permission from R.E. Rikli and C.J. Jones, 2013, pp. 89, 90.

Sit-and-Reach Test for Youth

This version of the sit-and-reach test focuses mainly on hamstring flexibility. The sit-and-reach test used in the FitnessGram for youth requires a 12-inch-high (30.5 cm) box and a yardstick (18). The following steps outline the setup and results:

- The yardstick is placed on the box with the zero end of the yardstick facing the child and the 9-inch (22.9 cm) mark at the nearest edge of the box. One foot is placed against the box and the other is flat on the floor next to the knee of the straight leg (see figure 7.3).
- The child reaches forward with back straight and head up. After measuring one side, have the child reverse the position of the legs and repeat.
- Record the number of inches for both the right and left sides to the nearest half inch (1.3 cm).
- Rather than determining a range, the test establishes a standard score to be met (or not). For boys ages 5 to 17, this is 8 inches (20.3 cm). For girls the standard increases with age: For 5- to 10-year olds, the standard is 9 inches (22.9 cm), for 11- to 14-year olds it is 10 inches (25.4 cm), and for 15- to 17-year olds it is 12 inches (30.5 cm).



FIGURE 7.3 Sit-and-reach for youth.

Shoulder Flexibility Assessments

Shoulder flexibility can affect a person's capability to perform activities of daily living such as brushing one's hair or reaching for a seat belt. The most common test for shoulder flexibility is the back-scratch test. Note that this test should not be attempted by anyone with neck or shoulder injuries or problems such as pinched nerves or frozen shoulder.

Back-Scratch Test for Adults

The only equipment you will need is a ruler. The following steps outline the setup and results (15):

- In a standing position, place your preferred hand behind the same-side shoulder, palm toward back and fingers extended, reaching down the middle of the back as far as possible (elbow pointed up).
- Then place your other hand behind the back, palm out, reaching up as far as possible in an attempt to touch or overlap the extended middle fingers of both hands (see figure 7.4). Do not attempt to grab your fingers and pull.
- Practice for two trials before measuring.
- The distance of overlap or distance between the tips of the middle fingers is measured to the nearest 1/4 inch. A minus score (–) is given to represent a distance short of touching; a plus score (+) represents the amount of an overlap.
- You can track your score over time. If hands can touch (or overlap), shoulder flexibility is a strong point, but if hands are short of reaching, shoulder flexibility is an area in which to seek improvement.



FIGURE 7.4 Back-scratch for adults.

Back-Scratch Test for Older Adults

This test requires only a ruler. The following steps outline the setup and results (16):

- In a standing position, place your right hand behind your right shoulder, palm toward back and fingers extended, reaching down the middle of the back as far as possible (elbow pointed up).
- Then place your left hand behind the back, palm out, reaching up as far as possible in an attempt to touch or overlap the extended middle fingers of your right hand (see figure 7.5).
- Practice for two trials and then repeat with your hands in the opposite position: left hand behind your left shoulder, reaching down toward your right hand that is placed behind your back.
- Whichever hand placement results in the closest reach should be used for scoring. Repeat the assessment twice with your hands in this position. Be sure to breathe normally and avoid bouncing or abrupt movements.
- The distance of overlap or distance between the tips of the middle fingers is measured to the nearest 1/2 inch. A minus score (–) is given to represent a distance short of touching (record the distance separating the middle fingers of each hand); a plus score (+) represents the amount of an overlap.
- Check your score in table 7.2. Normal ranges are shown; if your score is over the range listed, consider yourself above average and if your score falls short of the range listed, consider yourself below average.



FIGURE 7.5 Back-scratch for older adults.

TABLE 7.2 Normal Ranges for Back-Scratch Test for Older Adults in Inches

	Age						
	60 to 64	65 to 69	70 to 74	75 to 79	80 to 84	85 to 89	90 to 94
Males	–6.5 to 0.0	–7.5 to –1.0	–8.0 to –1.0	–9.0 to –2.0	–9.5 to –2.0	–9.5 to –3.0	–10.5 to –4.0
Females	–3.0 to +1.5	–3.5 to +1.5	–4.0 to +1.5	–5.0 to +0.5	–5.5 to +0.0	–7.0 to –1.0	–8.0 to –1.0

Adapted by permission from R.E. Rikli and C.J. Jones, 2013, pp. 89, 90.

Shoulder Stretch Test for Youth

No equipment is needed for this assessment. The objective is to touch the fingertips of opposite hands by reaching over the shoulder and under the elbow (18). The following steps outline the setup and results:

- Reach with the right hand, palm toward the back, over the right shoulder and down the back while at the same time placing the left hand, palm facing out, behind the back, and reach up toward the fingers of the right hand (see figure 7.6). Do the fingers touch?
- Then repeat with hands in the opposite position. Again, do the fingers touch?
- The scoring for this assessment is a simple “yes” or “no” response. The scoring is considered for both the right and left positions.



FIGURE 7.6 Shoulder stretch for youth.

Flexibility Program Components

Flexibility, like resistance training, is specific to the muscle groups and joints that are stretched. Thus, it is important to target all the major muscle groups (see figure 6.7, *a* and *b*, for the locations of the major muscle groups in the body).

A flexibility routine should be completed after a thorough warm-up of at least 5 minutes or after a cardiorespiratory or resistance training session. Increasing the temperature of the muscle increases its ability to stretch (1). Warm muscles have a greater elastic response than cold muscles do (3). The FITT-VP principle can be applied to your flexibility program, including the frequency, intensity, time, type, volume, and pattern of stretching activities.

Frequency

To improve flexibility, perform flexibility exercises at least two to three days per week for a minimum of 10 minutes (3). Note that this is considered a minimum; stretching on a daily basis as part of a warm-up or cool-down is effective in improving range of motion.



Flexibility does not have to reach this level to provide benefits.

Intensity

The question of how far to stretch (i.e., the intensity of the stretch) is a common one. Typically, stretching exercises are done to the point of mild tightness without discomfort within the range of motion of the joint(s) (3). If a given stretch creates discomfort, release slightly—a stretch should not be painful. Over time, you may be able to move the joint farther as your flexibility improves, but the stretch should never cause pain. If it does, back off slightly.

Time

You should hold a single flexibility exercise for 10 to 30 seconds (3, 5). In general, longer hold times have not been found to provide additional benefits for improving joint range of motion (3). However, older adults may benefit from holding the stretch for 30 to 60 seconds (3).

Type

Two of the most common methods of stretching to improve flexibility are static and dynamic. Both methods involve moving a joint or joints to the end of the range of motion. With static stretching, the position is held, whereas dynamic stretching involves continuous movement of the joint(s). Static stretching is more commonly used after an activity because some activities requiring strength, power, or endurance may be

Q&A

Should you perform static or dynamic stretches before or after a workout?

You may want to perform dynamic stretching before the workout, as these activities encourage large movements that raise the heart rate and increase blood flow to the muscles, tendons, and ligaments. Incorporating a dynamic warm-up has the potential to reduce injury as well as to prepare the body for the upcoming workout (12). But don't forget the static stretches following the workout. The musculoskeletal system is warm and ready for these lengthening exercises.

impaired by static stretching before the activity (3). Dynamic stretching can be done before activity, following a general warm-up of the muscles (20).

Static

Static stretching is undoubtedly the most common method used to improve flexibility. Static stretching consists of slowly moving a joint to the point at which you feel tension and then holding that stretch for 10 to 30 seconds (3). Remember, do not place your joints in any position that causes pain. As you hold the stretch, the tension should lessen as the muscle lengthens. Each static stretch should be repeated two to four times to accumulate 60 seconds per stretch.

Dynamic

Dynamic stretching involves moving parts of your body through a full range of motion while gradually increasing the reach and speed of the movement in a controlled manner. An example of this is arm circles; you begin with small, slow circles and gradually progress to larger and faster circles until you reach the full range of motion of the shoulder joint. Many people think dynamic stretching involves bouncing or jerking motions—it does not! The goal is to move the joint in a controlled manner within a normal range of motion in order to minimize the risk of injury (4). To avoid the muscle soreness that often results from novel movements, introduce dynamic stretches into your stretching program gradually, particularly if you are not accustomed to this type of stretching. Dynamic movements are typically repeated 5 to 12 times within a time frame that varies depending on the motion (approximately 30 to 60 seconds).

Volume and Pattern

In order to improve joint range of motion over time, a total of 60 seconds of flexibility exercise per joint is recommended (3). This is accomplished by repeating shorter-duration stretches, for example, repeating a 30-second stretch twice or repeating a 15-second stretch four times. Typically a body-wide stretching routine can be completed in less than 10 minutes per session (3).

Your Flexibility Program

Stretching can be done any time a muscle is warmed up and should be included before sports or activities requiring a high degree of flexibility. Stretching can be included

Flexibility Stretches to Avoid

Many stretches that have been accepted in the past have been found to cause unnecessary strain on the joints and muscles. Although not everyone engaging in these activities will incur injury, it is sensible to avoid certain stretches and focus on those included at the end of this chapter. Table 7.3 lists a few stretches to avoid and suggested alternatives.

Also note that in some situations, stretching a muscle may not be appropriate. For example, if a muscle or joint has been injured, stretching exercises would typically be postponed unless prescribed as part of a treatment plan by a health care provider.

TABLE 7.3 Stretches to Avoid and Suggested Alternatives

Stretch to avoid		Reason to avoid	Alternative stretch
Standing toe touch		May strain the lower back	Seated hamstring stretch (page 168)
Hurdler stretch		May put strain on the bent knee	Prone quadriceps stretch (page 169)
Overrounding of the back		May stress the neck and low back	Pillar–overhead reach (with slight torso rotation to involve trunk muscles) (page 163)
Hyperextension of the back		Ineffective at stretching the abdominal muscles and may put stress on the back	
Full neck circle		Hyperextends the neck	Forward and lateral flexion (page 169)

before or after the conditioning phase of general fitness activities. Although not conclusive, some research suggests that static stretching could interfere with sports that require muscular strength, power, or endurance (3). Thus, in the following sample programs, dynamic stretching follows the warm-up (before the conditioning phase of the workout), and static stretching is part of the cool-down.

Sample Stretching Program After a Warm-Up

After a thorough warm-up, dynamic stretches can be performed to improve the efficiency of the movements you will do during your conditioning period of cardiorespiratory or resistance training. Dynamic stretches should begin with small ranges of motion and progress to larger ranges of motion. You should repeat each movement 5 to 12 times or move continuously for 30 to 60 seconds. Figure 7.7 outlines a dynamic stretching program you can use after a warm-up.

Sample Stretching Program After a Conditioning Period

After a conditioning period, use static stretching to improve your flexibility. Figure 7.8 outlines a sample progressive static stretching program. When you begin a flexibility program, start with level 1 stretches, which are the most basic. Begin by holding static stretches for 10 seconds and slowly progress to holding the final position for up to 30 seconds, repeating each stretch two to four times. Once you are comfortable with level 1 stretches, progress to level 2 and then to level 3 stretches. The progression of certain stretching exercises (e.g., quadriceps) moves from a lying to seated to standing position. As you move through these levels, you will need more balance to perform the exercise.

If you are having trouble placing your body in the required positions, you can use a towel to provide some extension. For example, when doing the triceps stretch with

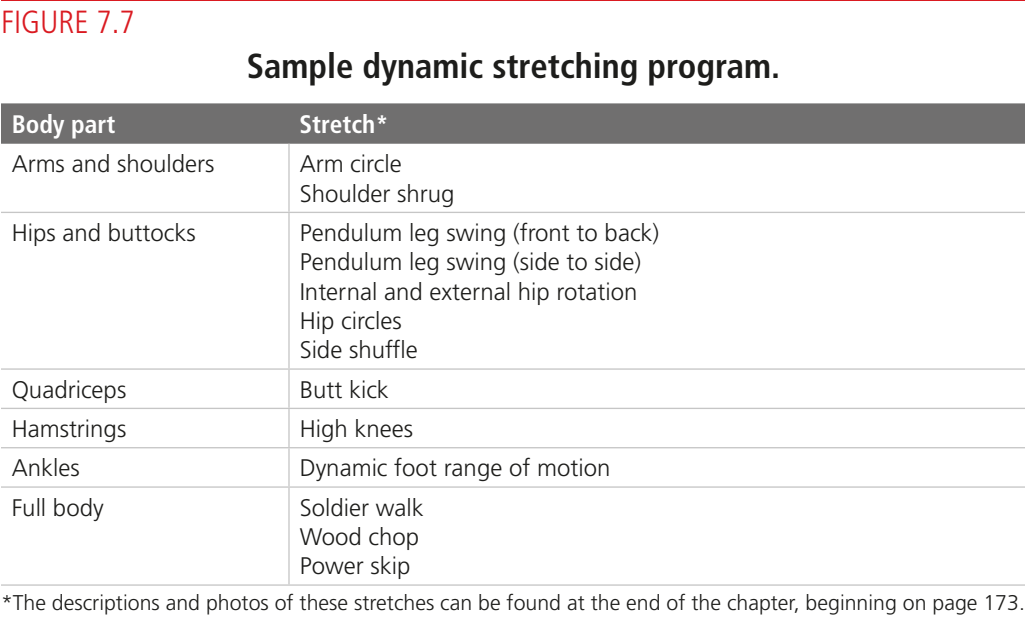


FIGURE 7.8

Sample progressive static stretching program.

Body part	Stretches by level of progression*		
	Level 1	Level 2	Level 3
Neck	Forward flexion Lateral flexion Levator scapulae stretch	Forward flexion Lateral flexion Levator scapulae stretch	Forward flexion Lateral flexion Levator scapulae stretch
Shoulders	Arms across chest	Arms across chest Wall hold	Arms across chest Wall hold
Upper back	Arm hug	Kneeling cat	Pillar–overhead reach
Low back	Supine rotational stretch	Supine rotational stretch	Supine rotational stretch
Chest	Chest stretch	Progressive chest stretch	Progressive chest stretch
Biceps	Biceps wall stretch	Biceps wall stretch	Biceps wall stretch
Triceps	Elbow behind the head	Elbow behind the head	Elbow behind the head
Hips and buttocks	Seated hip rotator stretch Butterfly stretch Kneeling hip flexor stretch	Supine hip rotator stretch Butterfly stretch Standing hip flexor stretch	Supine hip rotator stretch Butterfly stretch Standing hip flexor stretch
Hamstrings	Seated hamstring stretch	Standing hamstring stretch	Standing hamstring stretch
Quadriceps	Prone quadriceps stretch	Side-lying quadriceps stretch	Standing quadriceps stretch
Calves	Seated calf stretch	Standing calf step stretch (gastrocnemius) Standing calf stretch (soleus)	Standing calf step stretch (gastrocnemius) Standing calf stretch (soleus)

*The descriptions and photos of these stretches can be found at the end of the chapter.

the elbow behind the head later in this chapter, you could hold a towel in the hand of the arm you are stretching and provide assistance with the stretch by gently pulling on the towel with the other hand placed behind your back rather than on the elbow. When using a stretching aid, be careful not to jerk or pull your limb into an awkward or painful position.

Flexibility Stretches

The exercises to improve flexibility that have been listed throughout this chapter are provided here, organized by type—either static or dynamic. Each stretch includes a description and photos to help you perform it correctly.

STATIC STRETCHES

Static stretches, as discussed in detail previously in this chapter, are simple exercises that you can use to improve your flexibility. Remember to always warm up before stretching.

Neck

Forward flexion: Facing forward, move your head forward to tuck your chin into your chest; hold.



Lateral flexion: Facing forward, allow your head to tilt to the side so your ear moves toward your shoulder; hold. Repeat on the other side.



Levator scapulae stretch: Sit up straight on a chair. Put your hand up over your shoulder and bring your elbow back, pointing your elbow up to the ceiling. Use your left hand to pull your head forward and to the left; hold. Repeat on other side (6).



Shoulders

Arms across chest: Facing forward, straighten your right arm and draw it across your chest. Your arm should be as straight as possible, and you should feel gentle tension in your right shoulder. Grasp your right arm with your left hand and apply gentle pressure with your left hand to increase the tension in your right shoulder. Repeat on the other side.



Wall hold: Stand with your right side facing a wall. Place your hand on the wall at shoulder height with elbow straight and thumb pointing down (a). Turn your body away from the wall and maintain the rotation of your arm; hold (b). Repeat on the other side.



Upper Back

Arm hug: Cross your arms around your body with your elbows pointing forward. Let your upper body round, and squeeze your arms toward each other.



Kneeling cat: Adopt a crawl position on your hands and knees (a). Draw in your abdominal muscles and contract your buttocks, and then round your spine throughout its entire length (b).



Pillar–overhead reach: Facing forward, stand upright and extend your arms above your head, keeping your shoulders in a neutral position (in line with your hips). Interlock your fingers and use your palms to press upward. You can also involve your trunk muscles (torso) by slightly rotating to one side of your body and back. Hold when you feel tension in your torso on the side opposite the reach; repeat on other side.



Low Back

Supine rotational stretch: Lie face up on the floor and bend your knees so that your feet are flat on the floor. Straighten your arms out from your sides across the floor to stabilize your upper body (a). Slowly move both legs with your knees bent to the right side of your body while keeping your upper back against the floor and your abdomen oriented toward the ceiling (b). Repeat by moving your legs to the left side.



Chest

Chest stretch: In this stretch, your shoulders should be relaxed, not elevated. Straighten your arms toward your back, keeping them at or a little below shoulder height. A good cue for this stretch is "open arms wide."



Progressive chest stretch: Place your arms against an open doorway and lean forward until you feel gentle tension across your chest. This exercise also stretches the biceps.



Biceps

Biceps wall stretch: Position your arm from your hand to your inner elbow against a wall and turn your body away from it, exhaling slowly. Repeat on the other side.



Triceps

Elbow behind the head: Facing forward, bring your right arm up, bend from your elbow, and drop your hand behind your head, trying to reach your left shoulder with your right hand. The left hand can be placed on the right elbow to assist with this stretch. Repeat on the other side.



Hips and Buttocks

Seated hip rotator stretch: Sit upright on a sturdy chair that won't move. Cross your right ankle onto your bent left knee (a) and gently press down on your right knee until tension develops in the outer portion of your right thigh (b). Repeat on the other side.



Supine hip rotator stretch: Lie faceup on floor with your knees bent so your feet are flat on the floor and cross your right ankle onto your bent left knee (a). Lift your left foot off the floor and wrap your hands around your left leg and draw it into your body (b). Focus on opening up your right knee until tension develops in the outer portion of your right thigh. Repeat on the other side.



Butterfly stretch: Sit upright on the floor with the soles of your feet together. Draw your knees to the floor and lean forward from your hips and use your elbows to press your legs downward.



Kneeling hip flexor stretch: Kneel on both knees with your upper body lifted. Plant your left foot on the floor until you reach a 90-degree angle with both your front and back legs (a). Shift your weight forward while keeping your upper body lifted (b). Repeat on the other side.



Standing hip flexor stretch: Stand erect and keep your hands on your hips. Step forward with your left foot into a lunge position (a). Your left foot will be in front of your body and your right foot will be behind your body; your right heel may be elevated to facilitate this movement. Shift your hips forward and maintain this position, feeling tension develop in your hips, quadriceps, and buttocks (b). Repeat on the other side.



Hamstrings

Seated hamstring stretch: Sit upright on the floor with both legs straight and hands resting on your legs (a). Slowly walk your hands forward toward your feet, keeping your chest lifted (b).



Standing hamstring stretch: Standing upright, bring your right foot slightly ahead of your left foot. Slowly draw your hips back while slightly bending your left knee and straightening your right knee (a). Bring the toes of your right foot off the floor and toward your body (b). Hold and then return to the starting position. Repeat with the other leg.



Quadriceps

Prone quadriceps stretch: Lie facedown on the floor with your legs straight. Draw your right heel back toward your buttocks using your left hand. Be sure to keep your knees together.



Side-lying quadriceps stretch: Lie on the floor on your right side. Bend your left knee, keeping your knees and hips in a straight line (keep your knees together and do not twist your leg to the side). Draw your left heel back toward your buttocks with your left arm. Repeat on the other side.



Standing quadriceps stretch: While in a standing position (you can hold on to a chair for support), bend your right knee toward your buttocks. Grasp your right ankle with your left hand. Be sure to keep your knees close together and your ankle behind your buttock; do not twist your leg outward. Gently pull your thigh back slightly. Repeat on the other side.



Calves

Seated calf stretch: Sit upright with both legs straightened out in front of you (a). Draw your toes toward your upper body (b).



Standing calf step stretch: Stand with your legs extended on the edge of an immovable step and grasp a banister or handrail for support. Move your right foot so your heel back is off the edge of the step (a). Slowly drop your right heel until tension develops in your right calf (b). Repeat on the other side.



Standing calf stretch (gastrocnemius): Stand about 3 feet (0.9 m) from a wall and put your right foot behind you, ensuring that your toes are facing forward. Keep your heel on the ground and lean forward with your right knee straight. Rotating the toes in and out slightly will target the medial and lateral parts of this muscle separately. Repeat on other side.



Standing calf stretch (soleus): Stand away from a wall and put your right foot behind you and be sure your toes are facing forward. Lean forward at the ankle while bending the right knee and keeping your heel on the ground. Because the knee is flexed, tension is taken off the gastrocnemius and placed on the soleus. Repeat on other side.

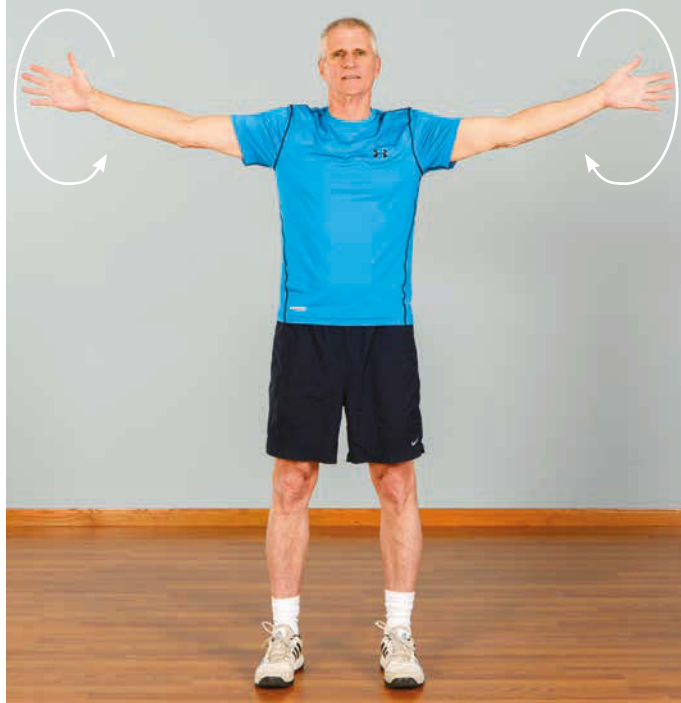


DYNAMIC STRETCHES

Dynamic stretches, as discussed in detail previously in this chapter, are more active than static stretches. Remember to always warm up before any stretching activity.

Arms and Shoulders

Arm circle: Stand with your feet shoulder-width apart and your knees slightly bent. Raise both arms to the side at shoulder height with your palms out. Make small circles with your arms extended, gradually increasing the size of the circles.

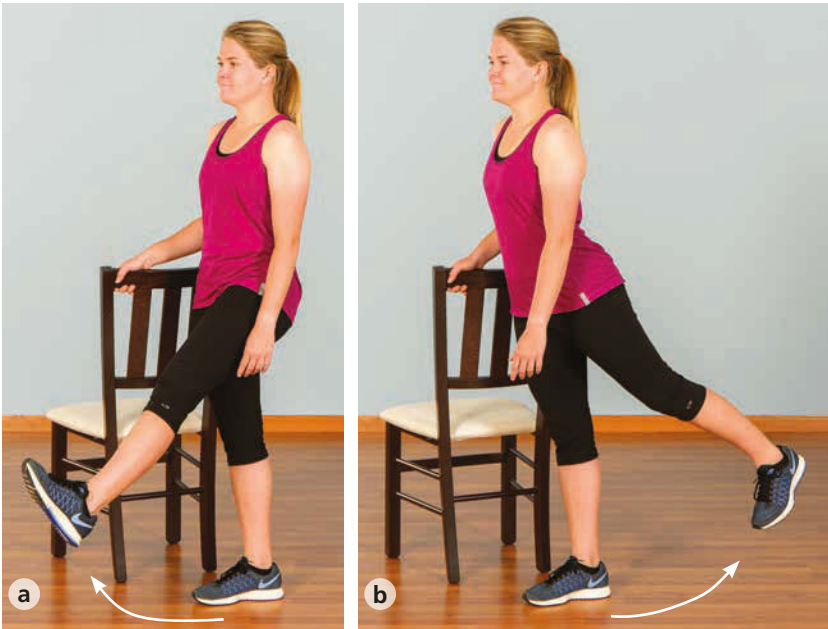


Shoulder shrug: Lift both shoulders toward your ears (a) and then lower them away from your ears (b).



Hips and Buttocks

Pendulum leg swing (front to back): Place your right hand on the back of a chair for balance. Lift your left leg and swing it forward (in front of your body) (a) and back (behind your body) (b). Begin with small swings and progress to larger swings. Switch to the opposite leg.



Pendulum leg swing (side to side): Place both hands on the back of a chair for balance. Swing your left leg out to the left (a), and back across your body to the right (b). Begin with small swings and progress to larger swings. Switch to the opposite leg.



Internal hip rotation: Stand upright with your feet shoulder-width apart. Raise your left foot toward the side of your body and tap the outside of your left heel with your left hand. Allow your knee to rotate inward. Switch and tap the outside of your right heel with your right hand. Alternate tapping each foot. Progress to walking forward while alternating feet.



External hip rotation: Stand upright with your feet shoulder-width apart. Raise your left foot in front of your body and tap the inside of your left heel with your right hand. Allow your knee to point away from your body. Switch and tap the inside of your right heel with your left hand. Alternate tapping each foot. Progress to walking forward while alternating feet.



Hip circles: Place your hands on your hips and feet spread wider than your shoulders. Make circles with your hips in a clockwise direction for 6 to 10 repetitions. Then repeat in a counterclockwise direction.



Side shuffle: Stand with your feet shoulder-width apart, your knees slightly bent, and your hands on your hips. Take one step to the left with your left foot (a); then bring your right foot in to meet your left foot (b). Begin with small steps, progress to larger steps, and then progress to a shuffle. Switch to the opposite direction.



Quadriceps

Butt kick: Begin marching in place. Pull your heel in closer toward your buttock with each step. Progress to moving forward (walking or jogging) while kicking your buttocks.



Hamstrings

High knees: Begin marching in place. Raise your knees higher and higher with each step. Progress to moving forward (walking or jogging) with high knees.



Ankles

Dynamic foot range of motion: Sit upright in a chair with both legs together and straightened in front of you. Point your toes away from your body and pull your toes toward your body (a). Rotate your feet clockwise and counterclockwise (b).



Combined Movements

Soldier walk: Simultaneously rotate your right arm forward and raise your left leg (straight). Reach your right hand toward your left lower leg and toes. Switch to the opposite side. Progress to alternating to the opposite side and then to walking while alternating sides.



Wood chop: Stand with your feet wider than shoulder width. Reach both arms down toward the outside of your left foot while bending your knees slightly (a). Move your arms diagonally across your body and end by reaching above your right shoulder (b). Switch to the opposite side.



Power skip: Skip across the field using powerful explosive movements. Use big arm swings starting from the side of the body through the frontal plane and reaching for the sky. Use high knee lifts moving the opposite arm (i.e., as the left leg moves forward, the right arm reaches upward).



Stretching exercises are recommended as an essential component of any exercise training program due to the improvement in range of motion and physical functioning. Improving flexibility can be accomplished through various stretching techniques, for example static or dynamic methods. Stretching exercises can be incorporated into the warm-up to help prepare the body for more vigorous activity, or following the conditioning period of a workout to enhance flexibility. And while the research in flexibility training is still emerging, following the guidelines set forth in this chapter will help you to improve your flexibility in a safe and effective manner.



EIGHT

Sharpening Your Functional Fitness

The importance of training specific body systems to improve health, fitness, and function has been discussed in the previous chapters. When trained properly, the cardiorespiratory and muscular systems provide individuals with the strength and stamina needed to perform a variety of simple and complex activities ranging from sitting, standing, and stepping to skipping rope, walking down stairs, or even running a marathon. Historically, the emphasis of health and fitness programs has been on challenging the cardiorespiratory and muscular systems and improving aerobic capacity, muscular fitness, and flexibility. However, over the past few decades, the importance of training another essential body system known as the neuromuscular system has been established.

The neuromuscular system is a complex and interconnected network that links the brain, spinal cord, and extremity nerves with sensory receptors and muscles located throughout the body. The role of the neuromuscular system is to integrate sensory information and, based on this information, to coordinate the appropriate muscle actions needed to produce a desired movement. The relationship between the various components of the neuromuscular system is similar to that of a musical conductor and the musicians in an orchestra. The conductor (the brain and spinal centers) is charged with directing the musicians (the muscles) in order to perform a specific musical piece. The conductor communicates with the musicians and directs them on how and when to play their instruments so that the correct notes are played with sufficient clarity, pitch, precision, and tempo (sensory information). If conducted effectively, the musicians execute a highly complex and precisely orchestrated musical performance (the desired motor task). Like the conductor directing musicians, the neuromuscular system uses sensory cues to control the muscles' actions with sufficient precision, coordination, and speed.

The neuromuscular system coordinates every motor task completed throughout the day, and the amazing part is that the majority of these tasks are performed with little

to no conscious effort. Even a simple task like getting dressed requires coordinated muscle activity. While getting dressed, did you consider engaging the muscles of your trunk, hips, and legs as you leaned forward to put on your shoes? Most likely you did not put a lot of thought into engaging all the muscles that were needed to carry out this activity. You simply considered the task that needed to be completed, and the right muscles were activated at precisely the right time. This is your neuromuscular system at work, and it has the extraordinary job of coordinating every muscle action for every movement you perform throughout the day.

The neuromuscular system helps people navigate their surroundings efficiently, effectively, and safely. Whether you are training for sport or for general health, this system is essential to maintaining balance, agility, coordination, and body awareness. Improving neuromuscular function may significantly reduce the risk for future falls and some musculoskeletal injuries (2, 15). Muscular fitness, cardiorespiratory endurance, and flexibility are important for long-term health and fitness; however, it would be difficult or even impossible to coordinate the thousands of muscle actions required to perform activities such as standing, walking, running, or jumping without a fully functional neuromuscular system. Fortunately, like the other body systems, the neuromuscular system can be trained to help the body respond more rapidly and economically to the physical demands faced in everyday life. The most effective means of training the neuromuscular system requires a targeted exercise strategy, which for the purposes of this chapter is referred to generally as neuromotor training. Neuromotor training, sometimes also referred to as functional fitness or sensorimotor training, involves specific exercises that challenge the neuromuscular system and are aimed at improving balance, agility, coordination, reaction time, and proprioception. This chapter outlines some of the important health and fitness benefits that can be derived from neuromotor training and details useful training tips to help you develop a personalized neuromotor training program based on your individual goals and needs.

Health and Fitness Benefits of Neuromotor Training

All movement requires a specific sequence of muscle actions, and the neuromuscular system coordinates and produces these muscle actions based on information learned from previous movement experiences. From infancy to adulthood, your neuromuscular system is continuously learning, processing, and storing new pieces of movement-related information that can be recalled at any time to help coordinate future motor tasks. Aging, deconditioning, musculoskeletal injury, and various neurological injuries and conditions can negatively affect neuromuscular function and movement quality. Neuromuscular function may begin to decline after the age of 30, resulting in diminished coordination and muscle control (24). Fortunately, emerging evidence suggests that neuromotor training can be an effective strategy for improving various skill-related components of fitness and may positively affect the structure and function of key brain and spinal centers involved in movement (1, 4, 27). The benefits of neuromotor training have been examined in aging and athletic populations and have been reported to improve balance, muscle strength, and agility and to reduce the risk of falls and some lower limb injuries (6, 11, 13, 14).

A number of underlying mechanisms have been attributed to neuromotor training including improved speed and efficiency of muscle recruitment, enhanced muscle force production, and improved reaction time in response to changes in environ-

mental conditions and body position (12). In addition, because of the highly dynamic and multidimensional nature of neuromotor training, it is likely that this type of training may induce greater changes in the nervous system, resulting in improved skill acquisition and retention, when compared to more stationary, one-dimensional exercises (2).

Multifaceted physical activities such as tai chi and yoga involve varying combinations of neuromotor, resistance, and flexibility exercise and have become popular training methods for individuals ranging from professional athletes to the aging population. Tai chi and yoga provide individuals with low-impact and relatively safe forms of neuromotor exercise that can directly benefit balance, motor control, and proprioception (11, 25). In recent years, the term *functional training* has become popular within health, fitness, and athletic training settings and is used to refer to a specific form of exercise training. Historically, functional training was used as a rehabilitation strategy to engage patients in exercises that closely resembled, if not entirely replicated, normal activities of daily living. Over the past decade, functional training, as a form of neuromotor training, has become a very popular training method. Although the exercises prescribed for athletes and healthy adults may require greater function and skill compared to those used in clinical settings, the principles of functional training, when used among healthy adults, still retain their clinical roots by basing exercise strategies on movement patterns that mimic activities in daily life or athletic competition.

A more detailed description of some of these activities is provided later in this chapter. Some of the possible benefits of functional training are improved agility, reaction time, muscle force production, and body control (3). Improvements in these areas can directly affect how well people react to changes in their environment, especially when faced with rapidly changing conditions such as those experienced when one trips, stumbles, or loses balance.

Despite the potential value of participating in neuromotor training activities in non-clinical settings, much still needs to be learned about the optimal duration, frequency, and intensity of training for long-term, sustainable health and fitness benefits. Definitive exercise recommendations for neuromotor training across all ages and ability levels have not been established; however, it is likely that benefits exist for anyone participating in physical activities that require agility, balance, and other motor skills or anyone who may be deficient in any of these areas (11).

Neuromotor Assessments

Similar to developing training programs for other components of fitness, it is helpful to first establish baseline measures of neuromotor function. There are a number of assessments that have been developed and that can be used to establish your starting point. These assessments range from very sophisticated laboratory measures to tests you can perform in your own home with minimal equipment. This section provides simple assessments you can perform at home or with a qualified exercise professional, including the 4-stage balance test, standing reach test, Edgren side-step test, agility T-test, and the 8-foot up and go (this is typically used for older adults only). Additional assessments within this book, such as the chair-stand test for muscular fitness included in chapter 6, may also be helpful for older adults (16). A selection of two or three tests should be sufficient to track functional neuromotor improvements over time, including balance, agility, coordination, and body awareness.

The 4-Stage Balance Test

This test assesses static balance and proprioception (5). You will need a stopwatch. The test includes four progressively more challenging standing positions. The following steps outline the setup and results:

- Stand near a wall or by a fixed object in the event that you lose balance and need to support yourself using your hands. Start by standing with shoes off and feet together (see figure 8.1a).
- Stand and hold this position for 10 seconds without holding on to anything for support. If you are able to maintain the position without losing your balance, shift your feet so the instep of one foot is touching the big toe of the other foot (see figure 8.1b).
- Stand and hold this position for 10 seconds without holding on to anything for support. If you are able to maintain the position without losing your balance, shift your feet into the tandem position in which your feet are placed heel-to-toe (see figure 8.1c). Repeat this until you are unable to balance for at least 10 seconds without moving your feet or needing to hold on to something for support.
- The goal is to advance to single-leg stance (see figure 8.1d).
- Once you are able to maintain the single-leg stance, use table 8.1 to determine your standing balance capacity (22). If you reach the “above average” range for your age group while standing with your eyes open, attempt the single-leg standing position with your eyes closed. Record the maximum time you can hold the single-leg stance position with eyes closed.



FIGURE 8.1 Four-stage balance test sequence.

TABLE 8.1 **Single-Leg Stance Time by Age Group**

	Age					
	18 to 39	40 to 49	50 to 59	60 to 69	70 to 79	>80
Eyes open (time in seconds)						
Above average	>43	>40	>37	>27	>15	>6
Average	43	40	37	27	15	6
Below average	<43	<40	<37	<27	<15	<6
Eyes closed (time in seconds)						
Above average	>9	>7	>5	>3	>2	>1
average	9	7	5	3	2	1
Below average	<9	<7	<5	<3	<2	<1

Adapted from B.A. Springer, R. Marin, T. Cyhan, et al., 2007, p. 11.

Standing Reach Test

This test assesses standing balance and postural control (7, 26). You will need a measuring stick and masking tape. The following steps outline the setup and results:

- Tape a leveled measuring stick on a wall horizontally at shoulder height.
- Stand with your right shoulder next to, but not touching, the measuring stick.
- Raise your arm to shoulder height (arm parallel to the ground), make a fist with your right hand, and note the number (in inches or centimeters) on the measuring stick that corresponds to the location of your knuckles (see figure 8.2a).
- When ready, with your arm outstretched, reach as far forward as you can without taking a step or losing balance (see figure 8.2b).

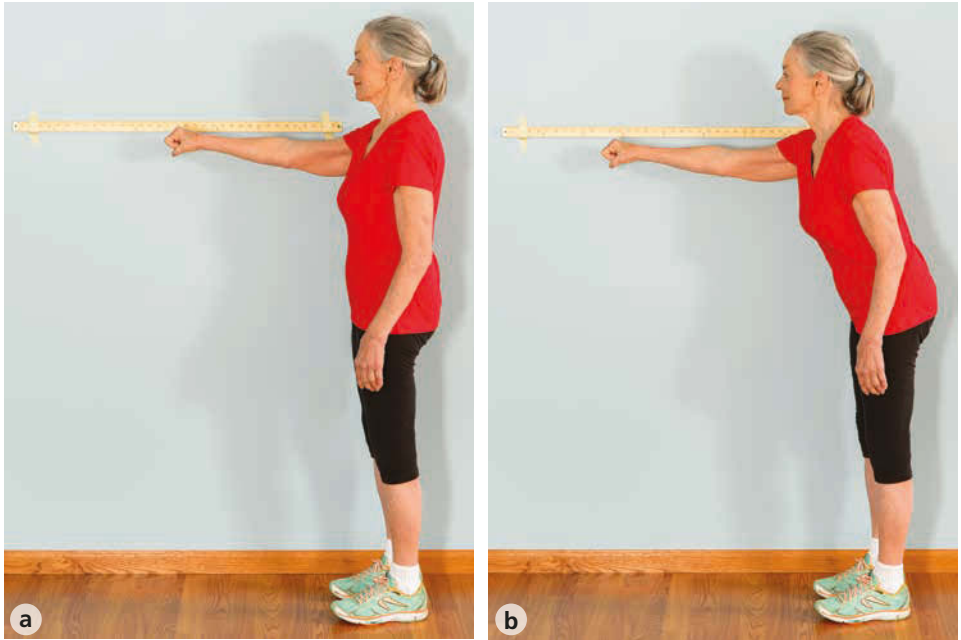


FIGURE 8.2 Standing reach test.

- Note the number (in inches or centimeters) on the measuring stick that corresponds to the location of your knuckles at the reaching position.
- Calculate the difference between the start and end reaching position.
- Use table 8.2 to determine your standing balance range (7). If you used a yardstick, convert your reach noted in inches to centimeters by multiplying by 2.54 (for example, a 6 inch reach would be 15 cm). A standing reach score less than 6 inches (15 cm) indicates a significant increased risk for falls, and a score of 6 to 10 inches (15-25 cm) indicates a moderate risk for falls.

TABLE 8.2 **Normal Ranges for the Standing Reach Test for Males and Females in cm**

Males	Age		
	20 to 40	41 to 69	>70 years
Above average	>19	>17	>15
Average	15 to 19	13 to 17	11 to 15
Below average	<15	<13	<11
Females	Age		
	20 to 40	41 to 69	>70 years
Above average	>17	>16	>15
Average	13 to 17	12 to 16	7 to 15
Below average	<13	<12	<7

Adapted from P.W. Duncan, D.K. Weiner, J. Chandler, and S. Studenski, 1990.

Edgren Side-Step Test

This test assesses sidestepping agility, quickness, and balance (8). You will need masking tape or cones, a tape measure, and a stopwatch. The following steps outline the setup and results:

- Find a flat, nonslip floor for your test location.
- Mark five lines using masking tape or place five cones in a line 3 feet (0.9 m) apart as illustrated in figure 8.3.
- Start the test standing at the center line or cone number 3.
- When ready, begin sidestepping to the right until your right foot touches or crosses the far right line or cone.
- Then, sidestep to the left until the left foot touches or crosses the far left line or cone.
- Continue to sidestep between lines or cones for 10 seconds and count the total number of lines or cones crossed when the test is complete.

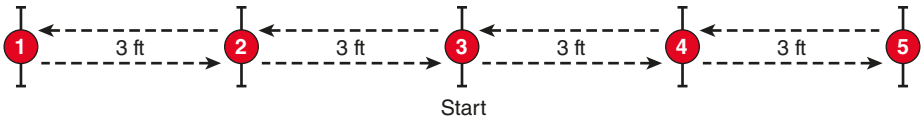


FIGURE 8.3 Edgren side-step test.

Adapted from H. Edgren, 1932.

- The test is scored based on the total number of line or cones crossed after 10 seconds. Since normal ranges are not available, use this assessment to track your score over time to see improvement. A better score is a higher number of lines or cones crossed during the 10-second period.

Agility T-Test

This test assesses agility in a forward, side, and backward direction (18). You will need a tape measure, cones, and a stopwatch. The following steps outline the setup and results:

- Set out four cones as illustrated in figure 8.4: 5 yards (4.6 m) and 10 yards (9.1 m) apart (21).
- Start at cone A.
- When ready, start the stopwatch and move as quickly as possible to cone B and touch the cone with your right hand.
- Then, sidestep left to cone C and touch the cone with your left hand.
- Then, sidestep to the right to cone D and touch the cone with your right hand.
- Sidestep to cone B, touch the cone with your left hand, and step backward to the start position at cone A. Stop the stopwatch and record the total time taken to complete the test.
- Use table 8.3 to compare results and track progress.

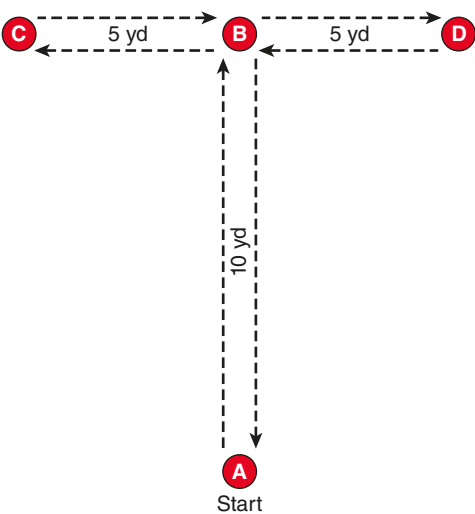


FIGURE 8.4 Agility T-test.
Adapted from K. Pauole, K. Madole, J. Garhammer, et al., 2000.

TABLE 8.3 Ranges for Agility Fitness in Males and Females in Seconds

	Males	Females
Excellent	<9.5	<10.5
Good	9.5 to 10.5	10.5 to 11.5
Average	10.5 to 11.5	11.5 to 12.5
Poor	>11.5	>12.5

Adapted from K. Pauole, K. Madole, J. Garhammer, et al., 2000.

8-Foot Up and Go Test for Older Adults

This test assesses agility and dynamic balance in older adults (19). You will need a standard-height chair (seat approximately 17 inches [43 cm] high) and a stopwatch. Steps to perform this test:

- Place the chair against a wall, and measure and mark a line 8 feet (2.4 m) away on the floor (see figure 8.5a).
- When ready, stand up from the chair.
- Walk to the line on the floor at a comfortable pace.
- Once both feet are past the line, turn around (see figure 8.5b).
- Walk back to the chair at a comfortable pace, and sit down again.
- Record the total time it takes you to stand from the chair, walk 8 feet (2.4 m), turn around, and sit back down.
- Normal ranges are found in table 8.4 (20). If your score is over the range listed, consider yourself above normal; if your score is below the range listed, consider yourself below normal. With this score, the shorter the time, the better (i.e., showing a faster completion of the test).

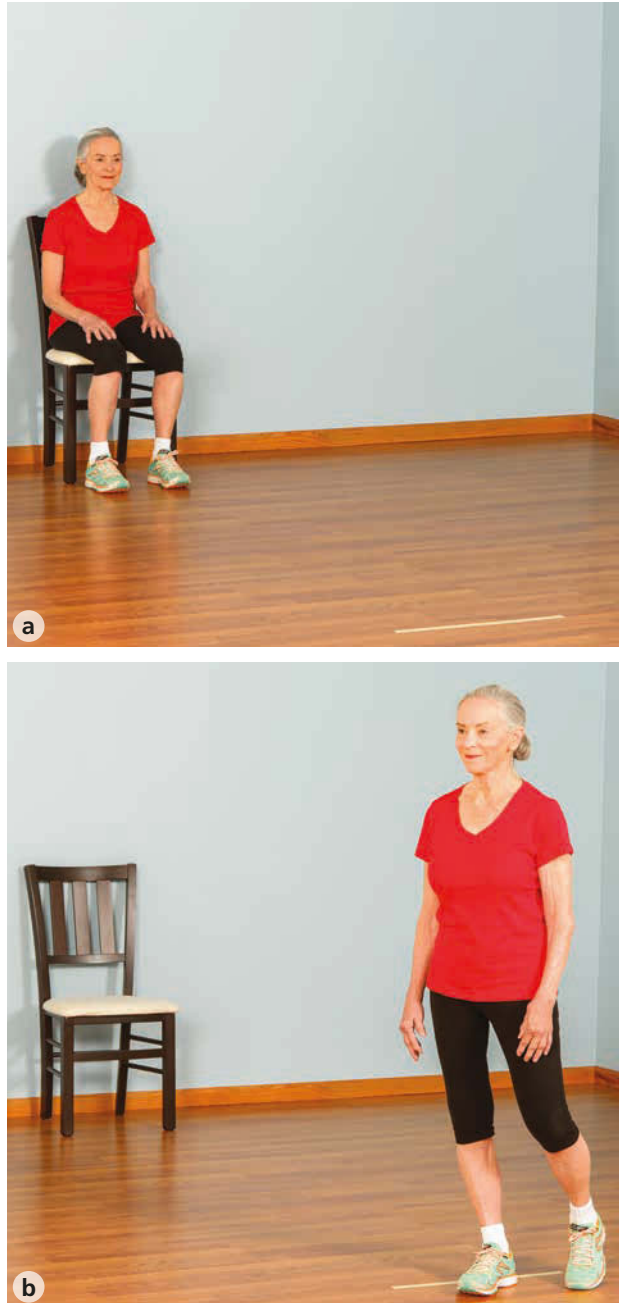


FIGURE 8.5 Eight-foot up and go for older adults.

TABLE 8.4 Normal Ranges for the 8-Foot Up and Go Test for Older Adults in Seconds

	Age						
	60 to 64	65 to 69	70 to 74	75 to 79	80 to 84	85 to 89	90 to 94
Males	3.8 to 5.6	4.3 to 5.9	4.4 to 6.2	4.6 to 7.2	5.2 to 7.6	5.5 to 8.9	6.2 to 10.0
Females	4.4 to 6.0	4.8 to 6.4	4.9 to 7.1	5.2 to 7.4	5.7 to 8.7	6.2 to 9.6	7.3 to 11.5

Adapted by permission from R.E. Rikli and C.J. Jones, 2013, pp. 89, 90.

Neuromotor Training Workout Components

Developing a safe and effective neuromotor training program requires consideration of the frequency, intensity, time, type, volume, and progression of exercises being performed. These training components vary depending on individual levels of physical conditioning, the presence of any preexisting injuries, and personal goals and needs. Because using exercise to specifically target the neuromuscular system in nonclinical populations is a relatively new training approach, there is no consensus concerning the optimal number of repetitions, intensity, or methods of progression for neuromotor exercise. This is due, in part, to the various types of training often included within neuromotor training. As a result, when research scientists and fitness professionals attempt to communicate the effectiveness of neuromotor training, it becomes difficult to identify whether improvements in health and function are associated with positive changes in neuromuscular function or are the result of improvements in some other body systems. However, there are some general exercise recommendations for frequency and duration that provide a good starting point for creating your own neuromotor training program (11). The general recommendations for neuromotor training presented in this chapter are based on the best and most recently available evidence.

Frequency

Neuromotor training exercise is recommended at least two to three days per week to improve balance and mobility (11). Note that this is only a suggested minimum; individuals who regularly participate in low-impact neuromotor training exercises such as tai chi, qigong, or yoga may be capable of performing these activities more frequently and may obtain additional health and fitness benefits without increasing the risk of injury (25). Neuromotor training exercises involving weighted resistance and explosive, high-impact activities (i.e., jumping, bounding, high-speed multidirectional agility) may place a greater physical stress on muscles, joints, and connective tissues. Under these training conditions, less frequent sessions of two or three days per week may be needed to allow for adequate recovery between sessions and to reduce risk of musculoskeletal injury. You may also consider fewer neuromotor training sessions if you are performing high-impact neuromotor exercises in conjunction with other forms of fitness training such as maximal strength training or high-volume aerobic training.

Intensity

The principle of overload states that in order to provide benefits from training, the intensity of exercise must be above and beyond that which is demanded of the body

on a day-to-day basis. To date, the intensity prescription for many neuromotor exercises, especially those targeting balance, has not been clearly established or adequately measured in research studies (9). Attempts at increasing neuromotor training intensity for the purposes of overloading the neuromuscular system have included increasing the duration of training and increasing the difficulty of the exercises (i.e., single- versus double-leg stance, narrow versus wide base of support, unstable versus stable surface) (9). The challenge is that the way in which people experience the intensity of balance, agility, coordination, and proprioceptive exercise can vary greatly. Monitoring movement quality may be helpful for assessing how demanding an activity is on your neuromuscular system. For example, if you are unable to maintain good form on any given exercise, then the exercise may be too advanced or your neuromuscular system may have become overwhelmed by the demands of the activity. In either case, if you are unable to maintain proper posture, body segment alignment, or balance while exercising, this may be a good indicator that your body has been challenged above and beyond its normal capabilities, and a short rest period may be needed before continuing.

Time

Current recommendations suggest that approximately 30 to 45 minutes should be devoted to neuromotor training for each session throughout the week (17). This should provide you with enough time to perform between 6 and 10 exercises depending on the demands of the specific activities you choose. Keep in mind that the neuromuscular system responds best to high-quality repetitive movements, so as your training progresses you may need to increase your training time as long as movement quality and body control are not compromised.

Type

Because the neuromuscular system is so heavily involved in the body's capacity to learn new activities, the principle of specificity may be one of the most important components to consider when developing a neuromotor training program. To illustrate this point, consider the task of learning to ride a bicycle. Riding a bicycle requires the development and coordination of a specific set of skills. You may have used or heard the saying "It's like riding a bike." This comparison reflects the neuromuscular system's ability to adapt to the specific demands of an activity and to easily recall motor skills related to that activity at a later time. It may take many hours or even days to develop the skills needed to effectively ride a bicycle. Yet the more you challenge your neuromuscular system, the more proficient your body becomes at the task of riding. Eventually, your neuromuscular system commits to memory the specific muscle actions needed to pedal, balance, and steer; and what started off as a challenging activity becomes very easy. The neuromuscular system is so proficient at learning and retaining information that even after many months or years have passed, you can climb back onto a bicycle and begin riding as if no time had passed at all.

Adhering to the specificity principle is critical to the development of an effective individualized neuromotor training program. Improvements in neuromotor function are specific to the types of activities you perform. If you want to reduce your risk for falling, then you must perform activities that challenge your upright stability and balance. If your goal is to improve coordination and agility for athletic competition,

then your training program must include sport-specific activities that challenge your neuromuscular system in this way. Lower extremity muscle strength can be improved through performance of repeated bouts of the seated leg press; however, improvements in seated leg strength may not translate to improved athletic performance if the neuromuscular demands of seated exercise are dramatically different from those experienced while evading tackles on the football field. Consequently, a multicomponent program involving task-specific neuromotor exercise may provide greater functional and performance benefit than one-dimensional exercise programs that focus on individual components of muscular strength, aerobic fitness, and flexibility (23). In addition, those forms of training that use various movements with and without visual feedback may be the most beneficial for improving specific components of neuromotor function such as proprioception and body awareness (2).

Volume

One of the most important aspects of neuromotor training is ensuring that you perform each exercise with the best form and technique possible. Your neuromuscular system learns from your repeated movement patterns. If you consistently perform an exercise incorrectly or in a way that does not engage the appropriate muscles in the right sequence or pattern, you may run the risk of “wiring” your neuromuscular system with the wrong series of muscle recruitment strategies. If you are new to exercise, knowing your physical limits and recognizing how your body responds to fatigue may be a challenge. Consulting a qualified exercise professional, even if only for a few sessions, may be helpful to guide you through proper exercise technique and form. This may better prepare you to recognize the signs of muscular fatigue and breakdown in movement performance and put you in a better position to optimize the benefits of your neuromotor training program.

Progression

Exercise progression and progressive overload are important concepts for all training. In order to maximize the potential benefits of neuromotor training, it is important to consistently and continuously challenge your neuromuscular system with activities that exceed the demands of your daily activities. For example, if sitting predominates in your day, then simple standing activities may be sufficient to challenge many neuromotor fitness domains. However, if your day involves significant time on your feet and possibly lifting, carrying, or moving objects, then it is likely that you will need to begin your neuromotor training program with more dynamic standing activities and possibly incorporate various standing surface conditions to optimize your benefits.

Although there is currently no clear consensus as to the most effective strategy for improving neuromuscular function through progressive neuromotor training, some logical progressions have been proposed. These progressions can be employed to ensure that your neuromotor training program effectively challenges your balance, coordination, agility, and proprioception (10). Table 8.5 provides a few examples of ways in which your neuromotor training program can be progressed through increasing degrees of difficulty. You can advance your neuromotor training program in almost an infinite number of ways, and no one way is necessarily better than another.

Progression of your exercise program will be based on your baseline level of physical conditioning and your personal comfort with performing different neuromotor exercises.

TABLE 8.5 **Sample Neuromotor Training Progressions**

Description	Sample levels of progression		
	Level 1	Level 2	Level 3
Change foot position and posture to gradually reduce the base of support.	Stand with feet side by side	Stand with feet in full-tandem position (heel to toe)	Stand on one leg
Vary dynamic movements that challenge the center of gravity.	Tandem walking	Braided walking (one foot crosses in front and then behind the other)	Backward walking
Challenge postural muscle groups.	Stand with feet flat on the ground	Stand with only heels touching the ground	Stand with only toes touching the ground
Gradually reduce sensory input.*	Stand with eyes open	Stand with one eye closed	Stand with both eyes closed
Complete exercises on progressively unstable surfaces.	Stand on a firm floor	Stand on a foam pad	Stand on a balance board or Bosu
Gradually increase movement speed.	Complete 20 side steps in 20 sec	Complete 20 sidesteps in 15 sec	Complete 20 sidesteps in 10 sec
Add weighted resistance to challenge balance and stability.	Stand in tandem while holding a single dumbbell by your side	Stand on one foot while holding a single dumbbell by your side	Stand on one foot while reaching down to pick up a kettlebell off the floor

*Note: Altering sensory input (such as with eyes closed) should be performed only while one is participating in stationary exercises. For safety, agility and multidirectional exercises should always be conducted with eyes open.

For example, you may find it more difficult to perform dynamic tasks (such as sidestepping or braided walking) and therefore need to begin your training with less dynamic stationary standing exercises. Likewise, you may find that stationary standing activities on firm, flat ground are very easy and therefore would need to begin with more difficult neuromotor activities like balancing on one foot while standing on an unstable surface. The focus of progression is to select exercises and levels of difficulty based on activities that are challenging but can be completed safely without increasing your risk of injury.

Your Neuromotor Training Program

A sample program for various levels is provided in figure 8.6. The time frame for the activities increases as the program is advanced. Sample exercises for each of the areas are provided in table 8.6 with pictures and descriptions later in this chapter.

Neuromotor Training Exercises

The neuromotor exercises you might wish to incorporate into your personalized training program are almost limitless. Your neuromotor training plan should be based on your individual needs and goals. Although these exercises are a great starting point, it is always helpful to consult a qualified exercise professional if you have any questions or concerns about developing a neuromotor training plan that is right for you.

FIGURE 8.6

Sample neuromotor training program.

	Exercises	Time per exercise	Total session time	Number of days per week
Beginner	<i>You should remain at this stage until you feel comfortable performing your selected exercises with good form, body control, and balance.</i>			
	Perform eight exercises in a circuit for time. Select two exercises from each of the following areas: agility, stationary balance, push, pull.	15 to 30 sec	20 min	2
Intermediate	<i>You should remain at this stage until you feel comfortable performing your selected exercises with good form, body control, and balance.</i>			
	Perform eight exercises in a circuit for time. Select two exercises from each of the following areas: agility, dynamic balance, push, pull.	30 to 45 sec	20 to 30 min	2 to 3
Advanced	<i>At this stage, continue to incorporate the suggested neuromotor training progressions while ensuring that you maintain good form, body control, and balance.</i>			
	Perform 8 to 10 exercises in a circuit for time. Select two or three exercises from each of the following areas: agility, dynamic balance, push, pull.	45 to 60 sec	30 to 45 min	2 to 3

TABLE 8.6 Neuromotor Training Exercise Guide

Category	Exercise
Stationary balance	Single-leg standing balance (p. 194)
	Single-leg forward reach (p. 194)
	Semi-tandem standing with diagonal reach (p. 195)
	Tandem standing balance (p. 196)
Agility	T-drill (p. 196)
	Lateral side step (p. 197)
	Braided side step (p. 198)
	4-square agility (p. 199)
Push	Push-ups (various forms) (p. 107, 108, 135, 201)
	Prone plank (p. 143)
	Up-down prone plank (p. 200)
Pull	Rows (various forms) (p. 136-139)
	Push-up hold with dumbbell row (p. 201)
Dynamic balance	Step-over hurdle (p. 202)
	Lunge with forward reach (p. 203)
	Step-up with overhead reach (p. 204)

STATIONARY BALANCE EXERCISES

Your ability to maintain balance affects many routine aspects of daily life. These simple exercises provide a number of options to help improve balance.

Single-Leg Standing Balance

Stand near a wall or by a fixed object in the event that you lose balance and need to support yourself. Begin with both feet on the ground (*a*). When ready, lift one knee toward the ceiling in a marching position (*b*). Hold the lifted leg in the air with the upper thigh parallel to the ground. Hold this position, without holding on to anything for support if you can, for the desired time. Repeat on the other leg. Once you become proficient at standing on one leg without support for 30 to 45 seconds, increase the proprioceptive challenge by eliminating your sight perception by closing your eyes.



Single-Leg Forward Reach

Begin by standing with both feet on the ground; then lift your right foot in the air behind you (*a*). With your right foot in the air, slowly bend forward at the waist, maintaining your balance, and reach your right hand toward the ground while keeping the right foot in the air the entire time (*b*). Once the limits of your stability have been reached, stand back up while making sure to continue to balance on the left foot only. Repeat this movement for the desired time or number of repetitions. Alternate your feet, standing on the right foot and lifting the left foot in the air. Bend forward while reaching your left hand toward the ground, and then once again return to the starting standing position without placing the left foot on the ground.



Semi-Tandem Standing With Diagonal Reach

Stand in a semi-tandem position with your left foot forward. Reach both arms in the direction of the back leg (a) and then reach up and across the body toward the direction of the forward leg (b). Switch to the opposite side.



Tandem Standing Balance

Start in a standing position with feet together and arms by your side. Slowly step forward with your left foot so that you are standing heel to toe with your left foot in front. Hold this position for 30 to 45 seconds and then step back to the starting position with feet together. Repeat this same sequence but with the right foot forward and the left foot back. Repeat this exercise for the desired time or number of repetitions. You can advance this exercise by performing the same sequence with eyes closed.



AGILITY EXERCISES

Agility exercises are used to challenge your body's ability to move and respond to changes in direction.

T-Drill

Follow the instructions for the Agility T-Test assessment earlier in this chapter. You will set up four cones in a T-shaped configuration (see figure 8.7) and then move from one cone to another by stepping forward, sideways, and backward. Complete as many circuits as possible following this movement pattern for the desired length of time.

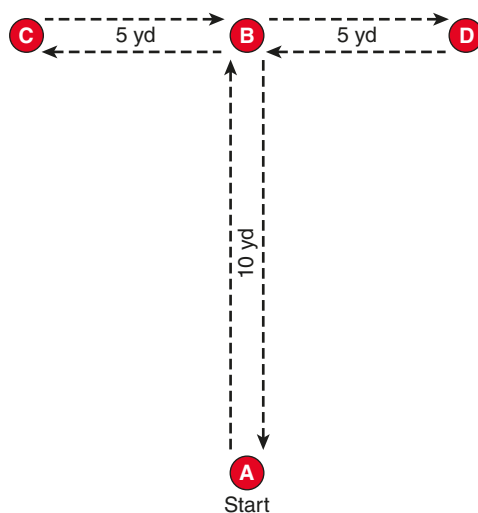


FIGURE 8.7

Adapted from K. Pauole, K. Madole, J. Garhammer, et al., 2000.

Lateral Side Step

Find a flat, nonslip floor for your exercise location. Mark two lines using masking tape or place two cones in a line, 15 feet (4.6 m) apart. You will sidestep toward your left (a) in order to touch or cross the line (b) and then repeat by sidestepping to your right (c). Continue to sidestep, without crossing your feet, as quickly as possible, but safely, between the lines or cones for the desired length of time.



Braided Side Step

For the braided side step, follow the setup and execution as described in the lateral side-step exercise; the only difference is that your foot positions vary as you move from one line or cone to the other. In the braided side step, one foot crosses in front and then behind the other as you move from side to side (*a* and *b*). Move your feet as quickly as possible while still maintaining balance and body control.



4-Square Agility

Begin by cutting two pieces of tape and placing them on the floor, one crossing the other through the center. Label the squares 1, 2, 3, and 4. Begin with both feet in square 1. When you are ready, work your way from square 1 to square 2 to square 4 to square 3 and back to square 1 by stepping with both feet to the side, backward, to the side, and forward, respectively (*a-d*). Complete this sequence 10 times consecutively. Make it your goal to complete each circuit as quickly as you possibly can but safely. To increase the challenge of this exercise, transition from stepping to jumping from square to square, making sure not to touch the lines in the center with your feet. Additionally, you can increase the challenge by varying the number sequence (e.g., 4-1-3-2, 4-2-3-1, 1-4-2-3, 1-3-4-2).



PUSH EXERCISES

The exercises in this section range from simple movements (e.g., push-ups) to more complex movements (e.g., up-down plank) that may be more appropriate once you have established a foundation of neuromotor training. Chapter 6 includes a number of these exercises (e.g., push-up options and prone plank).

Up-Down Prone Plank

This exercise combines and advances the full push-up and prone plank. Begin in the prone plank position (a). Shift your body weight to the left elbow and shoulder while simultaneously transitioning: moving from your right elbow to your right hand (b). Next, transition your weight to the right hand and shoulder, pushing up through the right arm and placing your left hand on the ground and pushing into full push-up position (c). Repeat the steps in reverse until you return to the prone plank position on your elbows. Repeat the sequence for the desired number of repetitions or time. Be sure to alternate sides on which you are pushing up from the prone plank position so as not to work one arm and shoulder more than the other.



PULL EXERCISES

In addition to push exercises, the opposing movement—pulling—should be included in your neuromotor training program. Chapter 6 presents a number of options such as row exercises. Another option including the push-up position with the row movement is described in this section.

Push-Up Hold With Dumbbell Row

Assume a full push-up position but do so with hands holding two dumbbells (*a*). Lower yourself down to the floor, and when the limits of your range of motion have been achieved, push up and lift one dumbbell off the floor toward your side (*b*). Slowly lower the dumbbell back to the floor. Repeat this process for the desired amount of time or number of repetitions, remembering to alternate dumbbell lifts between the left and right arms.



DYNAMIC BALANCE EXERCISES

In addition to the stationary balance exercises, dynamic balance exercises include the additional challenge of movement.

Step-Over Hurdle

Place a small box (e.g., shoe box) or taped line on the floor. Stand with your right shoulder facing the box (a). When ready, lift your right knee toward the ceiling and step over the box to your right (b). Place your right foot on the ground (c), and lift your left foot over the box so that your left shoulder is now facing the box (d and e). Repeat this exercise from side to side until the desired time or number of repetitions has been completed.



Lunge With Forward Reach

Begin in a standing position with both feet together and arms by your side (*a*). Take a stride-length step forward with the left foot. In the lunge position, bend both knees so that the back knee moves downward toward the floor. As your knees bend, ensure that your trunk remains upright with the right shoulder, hip, and knee aligned. Once in this position, reach your arms out in front of you in order to further challenge your balance and limits of stability (*b*). Once you have reached your maximum range of motion with the knees bent, push off the left foot, return to the start position with both feet together, and return your arms to your sides. Repeat this with your right foot stepping forward. Reach your arms out in front while maintaining your balance, and then return to the start position. Continue to alternate lunges between the left and right leg until the desired time or number of repetitions has been achieved.



Step-Up With Overhead Reach

Stand in front of a step (a). Begin by stepping up with the right foot (b). Push yourself into a standing position on the step as you reach your arms overhead to the limits of your shoulder range of motion (c). Step backward and down off the step, returning your arms to your side. Continue by stepping up with the left foot and reaching overhead as you come to a standing position on the step. Repeat until the desired time or number of repetitions has been reached. For an additional challenge when stepping up onto the bench with the right foot, keep the left foot elevated. Maintain this position with arms overhead for a moment before stepping back down with the left foot, followed by the right foot to return to the starting position. Continue, alternating feet.



Although researchers are still seeking to identify the optimal frequency, intensity, time, type, and progression of neuromotor exercise, one thing is clear: Whether one is training for sport or for general health, neuromotor training is a recognized and necessary component of a comprehensive exercise training program. In the coming years, with advances in research and professional practice, exercise professionals and the broader exercise community will gain a much better understanding of the important role that neuromotor training plays in helping individuals of all ages and ability levels maintain optimal health, fitness, function, and quality of life.